GLOMOS
GLOBAL MOUNTAIN SAFEGUARD RESEARCH

UNI GRAZ

Asian Mountain Academic Alliance

EBERHARD KARLS
UNIVERSITÄT TÜBINGEN

May 2026

climate risk in vulnerable mountain regions of the world

Day 3 (28 May)

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Course content

22 May (1400 – 1700)

Pre course evaluation - general introduction – the mathematics of risk – introduction to assignments

27 May (0900 – 1500)

Examples of hazards / exposure types /vulnerability assessments

Communicating risk

28 May (0900 – 1500)

International initiatives

Science to policy

29 May (0900 – 1500)

Serious game – course wrap up

Course content

28 May (0900 – 1500)

09:00 – 09:45	Disaster Risk Reduction
09:45 – 10:00	Break
10:00 – 10:45	Concrete recent examples
10:45 – 11:00	Break
11:00 – 12:00	International Platforms to address Mountain Risks
12:00 - 13:00	Lunch Break
13:00 - 13:45	Intro to the Serious Game – how do international deliberations look like?
13:45 - 14:00	Break
14:00 - 15:00	new data and method avenues

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Disaster Risk Reduction

Disaster risk reduction (DRR) is defined by United Nations Office for Disaster Risk Reduction (UNDRR) as those actions which aim to "prevent new and reducing existing disaster risk and managing residual risk, all of which contribute to strengthening resilience and therefore to the achievement of sustainable development".

UNDRR
UN Office for Disaster Risk Reduction

GFDRR
Global Facility for Disaster Reduction and Recovery

Disaster Risk Reduction – Sendai Framework

Chart of the Sendai Framework for Disaster Risk Reduction

2015-2030

Scope and purpose

The present framework will apply to the risk of small-scale and large-scale, frequent and infrequent, sudden and slow-onset disasters, caused by natural or manmade hazards as well as related environmental, technological and biological hazards and risks.
It aims to guide the multi-hazard management of disaster risk in development at all levels as well as within and across all sectors

Expected outcome

The substantial reduction of disaster risk and losses in lives, livelihoods and health and in the economic, physical, social, cultural and environmental assets of persons, businesses, communities and countries

Goal

Prevent new and reduce existing disaster risk through the implementation of integrated and inclusive economic, structural, legal, social, health, cultural, educational, environmental, technological, political and institutional measures that prevent and reduce hazard exposure and vulnerability to disaster, increase preparedness for response and recovery, and thus strengthen resilience

Targets

Substantially reduce global disaster mortality by 2030, aiming to lower average per 100,000 global mortality between 2020-2030 compared to 2005-2015	Substantially reduce the number of affected people globally by 2030, aiming to lower the average global figure per 100,000 between 2020-2030 compared to 2005-2015	Reduce direct disaster economic loss in relation to global gross domestic product (GDP) by 2030	Substantially reduce disaster damage to critical infrastructure and disruption of basic services, among them health and educational facilities, including through developing their resilience by 2030	Substantially increase the number of countries with national and local disaster risk reduction strategies by 2020	Substantially enhance international cooperation to developing countries through adequate and sustainable support to complement their national actions for implementation of this framework by 2030	Substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information and assessments to people by 2030

Priorities for Action

There is a need for focused action within and across sectors by States at local, national, regional and global levels in the following four priority areas.

Priority 1
Understanding disaster risk

Disaster risk management needs to be based on an understanding of disaster risk in all its dimensions of vulnerability, capacity, exposure of persons and assets, hazard characteristics and the environment

Priority 2
Strengthening disaster risk governance to manage disaster risk

Disaster risk governance at the national, regional and global levels is vital to the management of disaster risk reduction in all sectors and ensuring the coherence of national and local frameworks of laws, regulations and public policies that, by defining roles and responsibilities, guide, encourage and incentivize the public and private sectors to take action and address disaster risk

Priority 3
Investing in disaster risk reduction for resilience

Public and private investment in disaster risk prevention and reduction through structural and non-structural measures are essential to enhance the economic, social, health and cultural resilience of persons, communities, countries and their assets, as well as the environment. These can be drivers of innovation, growth and job creation. Such measures are cost-effective and instrumental to save lives, prevent and reduce losses and ensure effective recovery and rehabilitation

Priority 4
Enhancing disaster preparedness for effective response, and to «Build Back Better» in recovery, rehabilitation and reconstruction

Experience indicates that disaster preparedness needs to be strengthened for more effective response and ensure capacities are in place for effective recovery. Disasters have also demonstrated that the recovery, rehabilitation and reconstruction phase, which needs to be prepared ahead of the disaster, is an opportunity to «Build Back Better» through integrating disaster risk reduction measures. Women and persons with disabilities should publicly lead and promote gender-equitable and universally accessible approaches during the response and reconstruction phases

Guiding Principles

- Primary responsibility of States to prevent and reduce disaster risk, including through cooperation
- Shared responsibility between central Government and national authorities, sectors and stakeholders as appropriate to national circumstances
- Protection of persons and their assets while promoting and protecting all human rights including the right to development
- Engagement from all of society
- Full engagement of all State institutions of an executive and legislative nature at national and local levels
- Empowerment of local authorities and communities through resources, incentives and decision-making responsibilities as appropriate
- Decision-making to be inclusive and risk-informed while using a multi-hazard approach

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- Coherence of disaster risk reduction and sustainable development policies, plans, practices and mechanisms, across different sectors
- Accounting of local and specific characteristics of disaster risks when determining measures to reduce risk
- Addressing underlying risk factors cost-effectively through investment versus relying primarly on post-disaster response and recovery
- «Build Back Better» for preventing the creation of, and reducing existing, disaster risk
- The quality of global partnership and international cooperation to be effective, meaningful and strong
- Support from developed countries and partners to developing countries to be tailored according to needs and priorities as identified by them

Target A: Disaster-related deaths and missing persons (excluding COVID-19)

- Average disaster-related mortality per 100,000 population declined from 1.60 in 2005-2015 to 1.15 in 2013-2022 (a decline of 28%).
- During 2015-2022, the average annual disaster-related mortality stood at 41,789 persons in absolute terms.

Target B: Disaster-affected population

- Number of disaster-affected people Increased from 1,092 per 100,000 population during 2005-2015 to 2,034 during 2013-2022 (an increase of 86%).
- On an average, more than 130 million people were affected by disasters each year globally during 2015-2022.

Target C: Disaster-related economic losses

- Direct economic loss averages 0.37% of global GDP of reporting countries during 2015-2022.
- Provisional data shows that in 2022, direct economic loss accounts for 0.83% of global GDP.

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Target D: Disaster damage to critical infrastructure and disruption of basic services

- From 2015 to 2022, over 800,000 critical infrastructure units and facilities, including schools and hospitals, were damaged or destroyed due to disasters.
- In the same period, more than 12 million educational, health, and other basic services were disrupted by disasters.

Target E: National and local disaster risk reduction (DRR) strategies

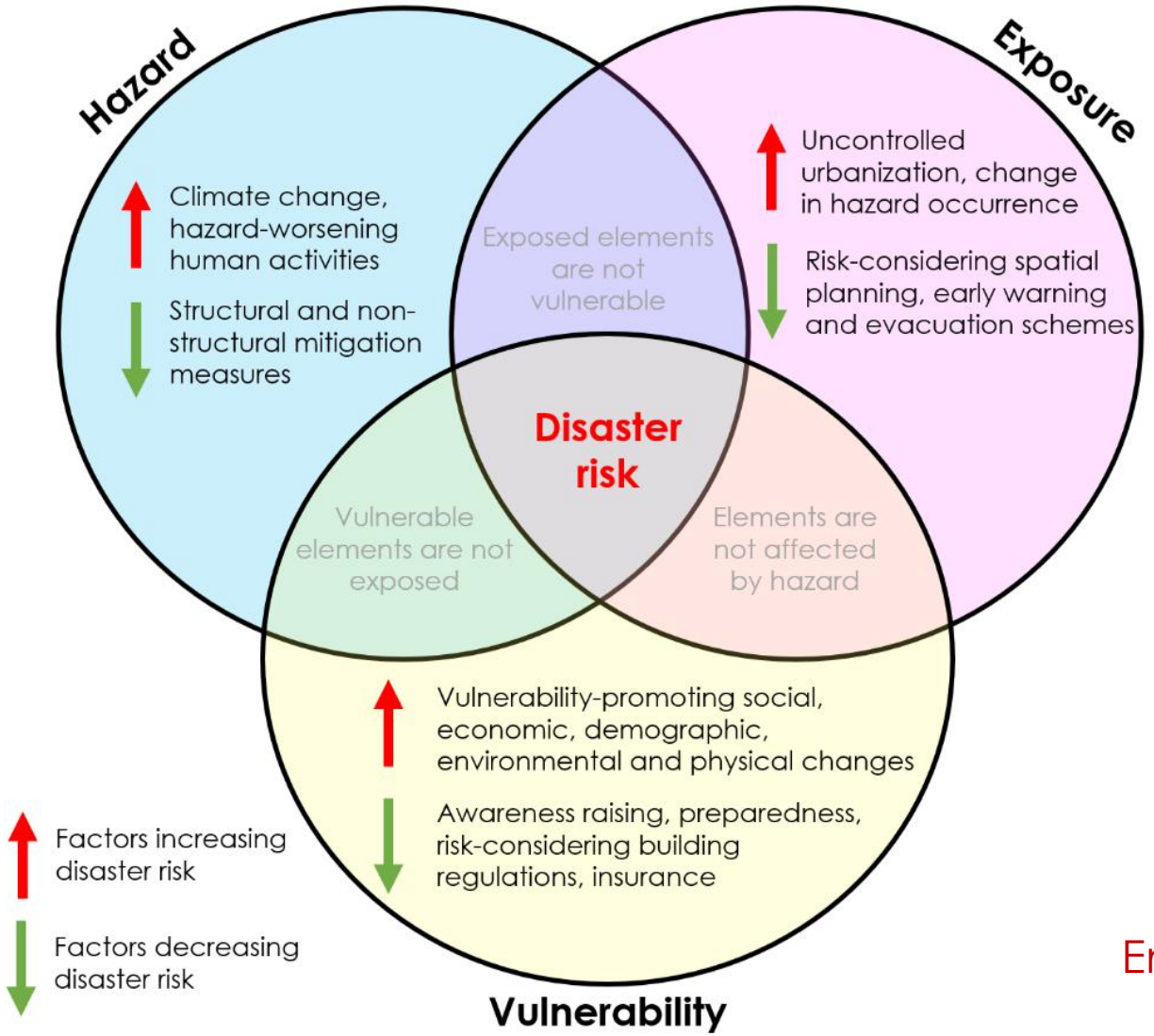
- 126 countries have reported having national disaster risk reduction strategies, compared to 55 in 2015 (two-third of countries in the world and an increase of 130% since the adoption of Sendai Framework).
- 102 countries have reported having local disaster risk reduction strategies. Average proportion of local governments in such countries is 72%.

Target F: International cooperation for disaster risk reduction

- International cooperation for developing countries remains limited, as developing countries reported only US$ 304 million of official development assistance (ODA) plus other official flows for supporting disaster risk reduction actions in 2021.
- From 2005 to 2022, 1,742 programmes and initiatives on science, technology and innovation transfer in disaster risk reduction were reported by 39 countries.
- In the same period, 33,441 instances of capacity development on disaster risk reduction were reported by 42 countries.

Target G: Multi-hazard early warning systems (MHEWS)

- 101 countries reported having MHEWS, compared to 47countries in 2015 (an increase of 114%).
- Two third (66%) of the countries reported having MHEWS have self-assessed their MHEWS with “limited to moderate achievement.”



Emmer, Steiner and Kalvoda (2024)

Risk management – disaster risk reduction

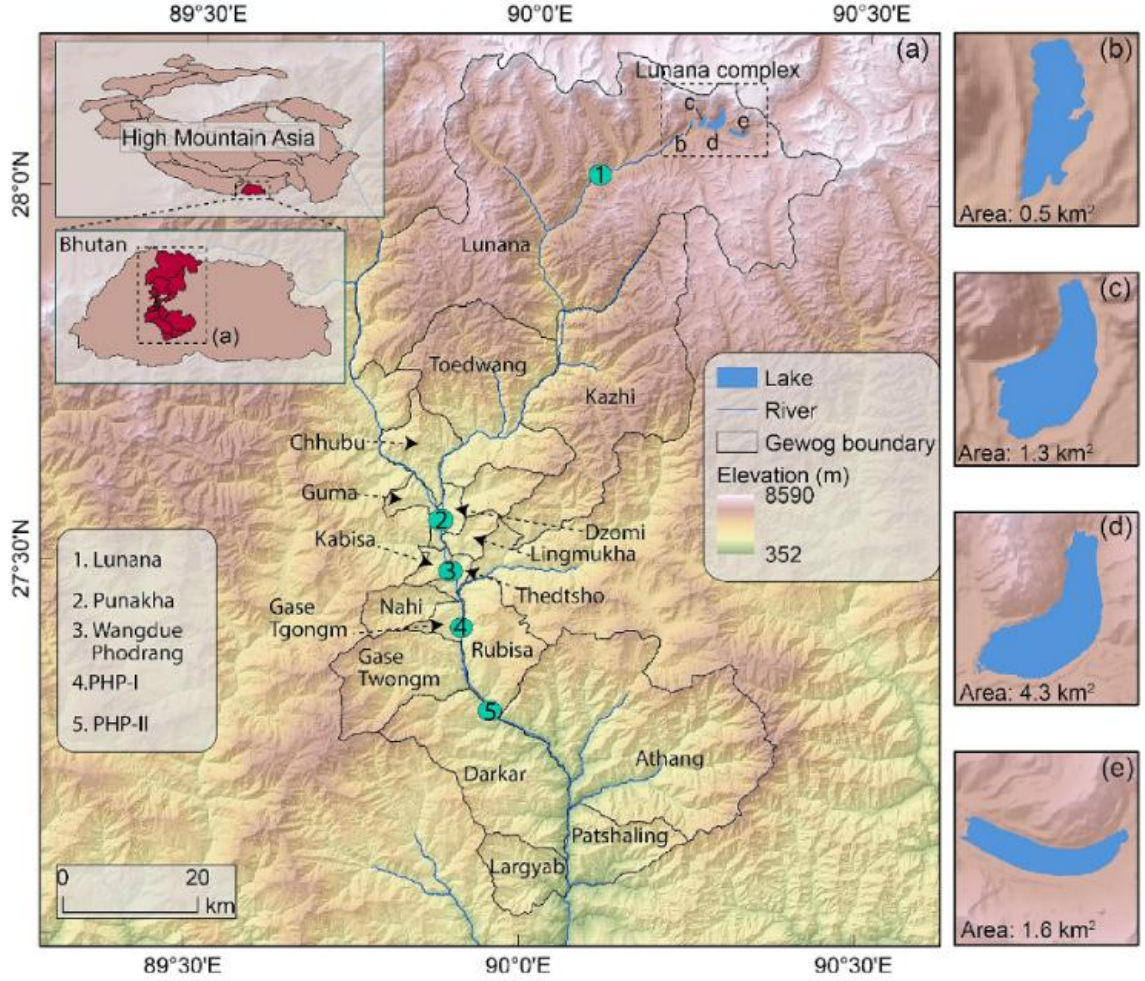


Fig. 1. Lunana glacier complex and downstream settlements. Map (a) shows settlements downstream of the Lunana glacier complex along the Phochu and Punatsangchu basins. Downstream settlements are presented following the subdistrict block administrative boundary. Panels (b–e) show four glacial lakes, which we studied here: Bechung Tsho, Raphstreng Tsho, Lugge Tsho, and Thorthormi Tsho, respectively. Background image: ALOS-PALSAR DEM.

Rinzin et al., 2023

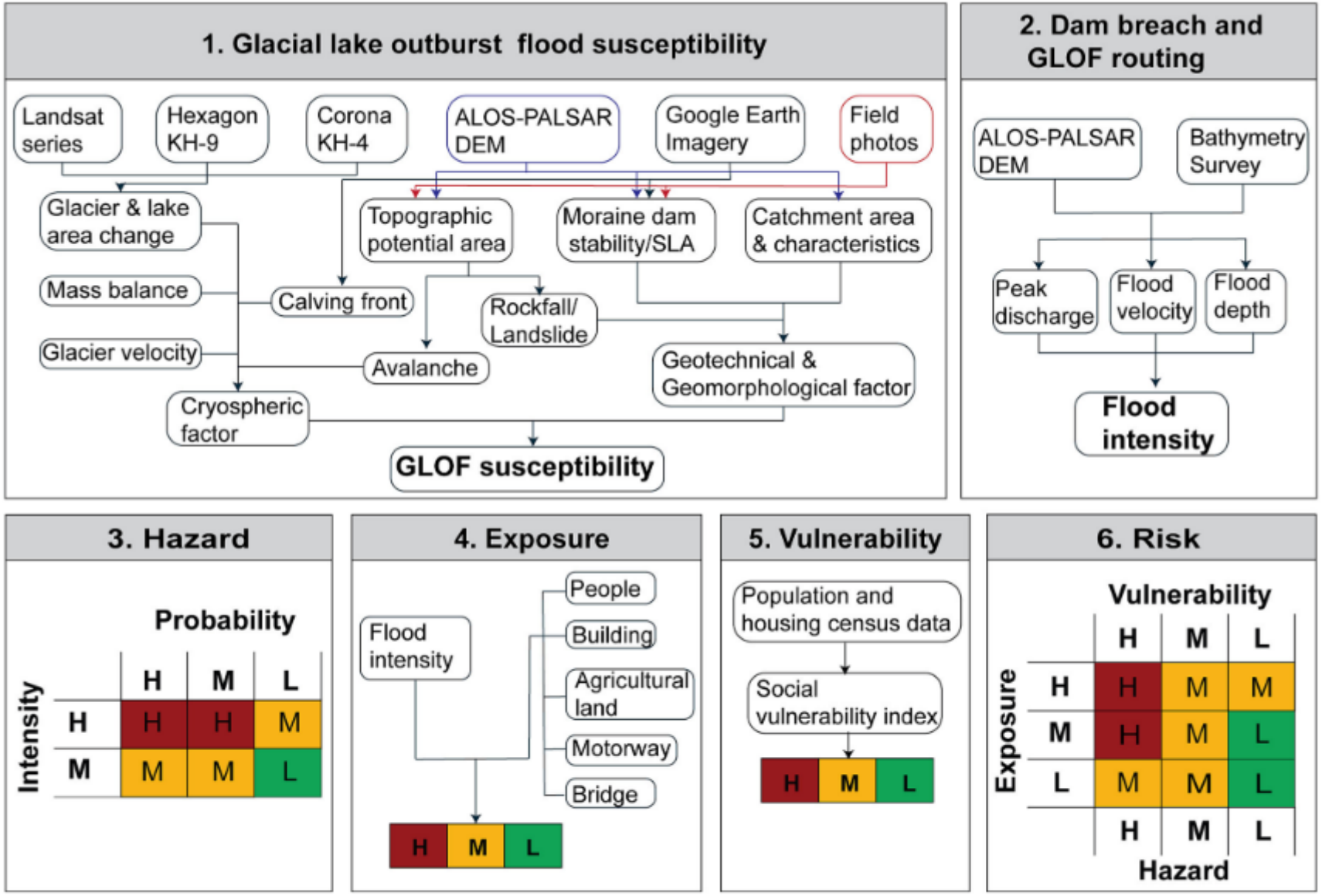


Fig. 2. Methodological flow chart adopted for GLOF susceptibility, downstream hazard, exposure, vulnerability, and risk assessments for the potentially dangerous glacial lakes in the Lunana glacier complex of the Bhutan Himalaya: downstream GLOF hazard, vulnerability, exposure, and risk are categorized into three classes; high (H), medium (M) and low (L). For the hazard computation, the GLOF intensity used here is translated from the modelled flood depth: high (more than 1 m) and medium (less than 1 m) (after GAPHAZ, 2017).

Rinzin et al., 2023

Hazards

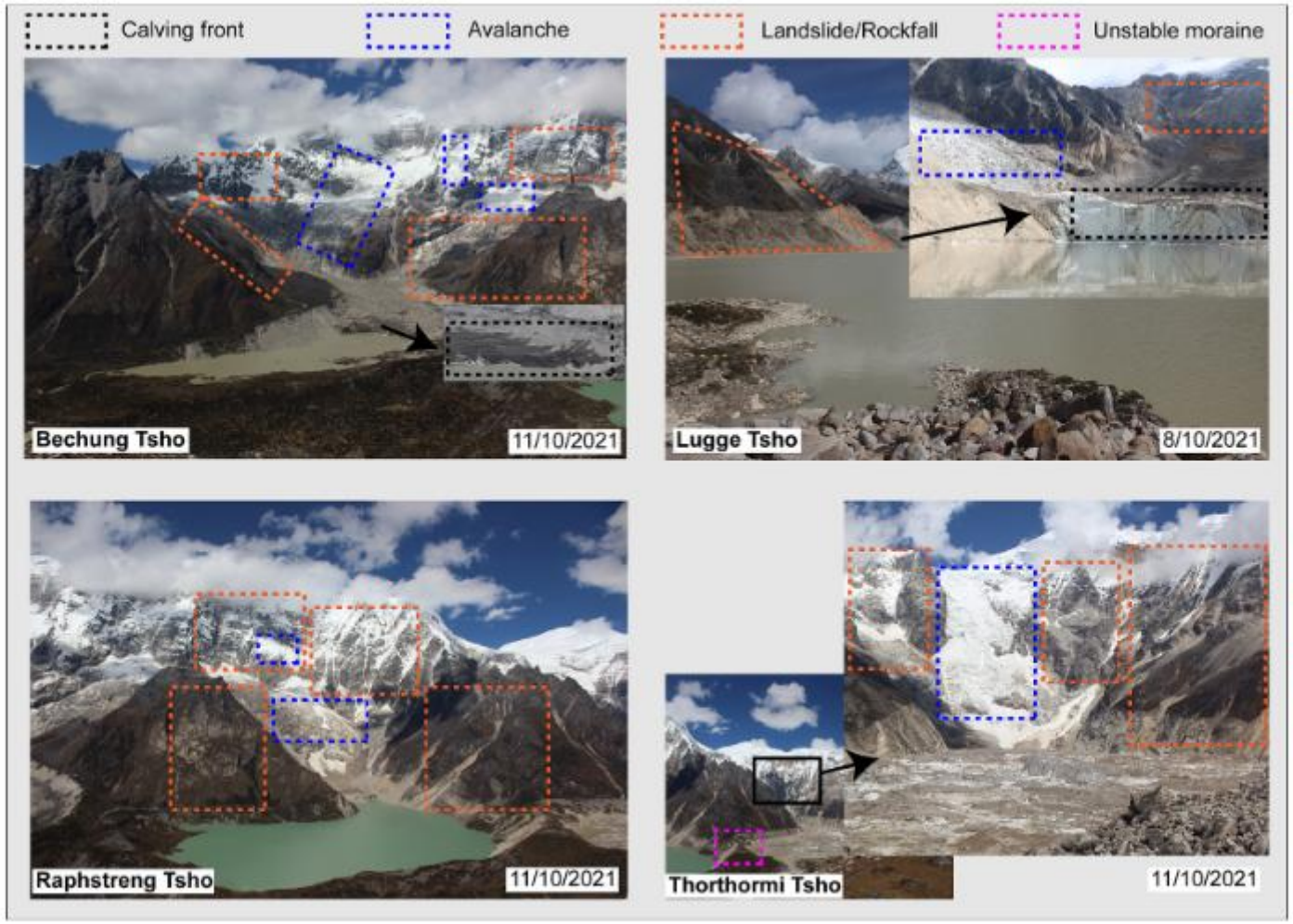
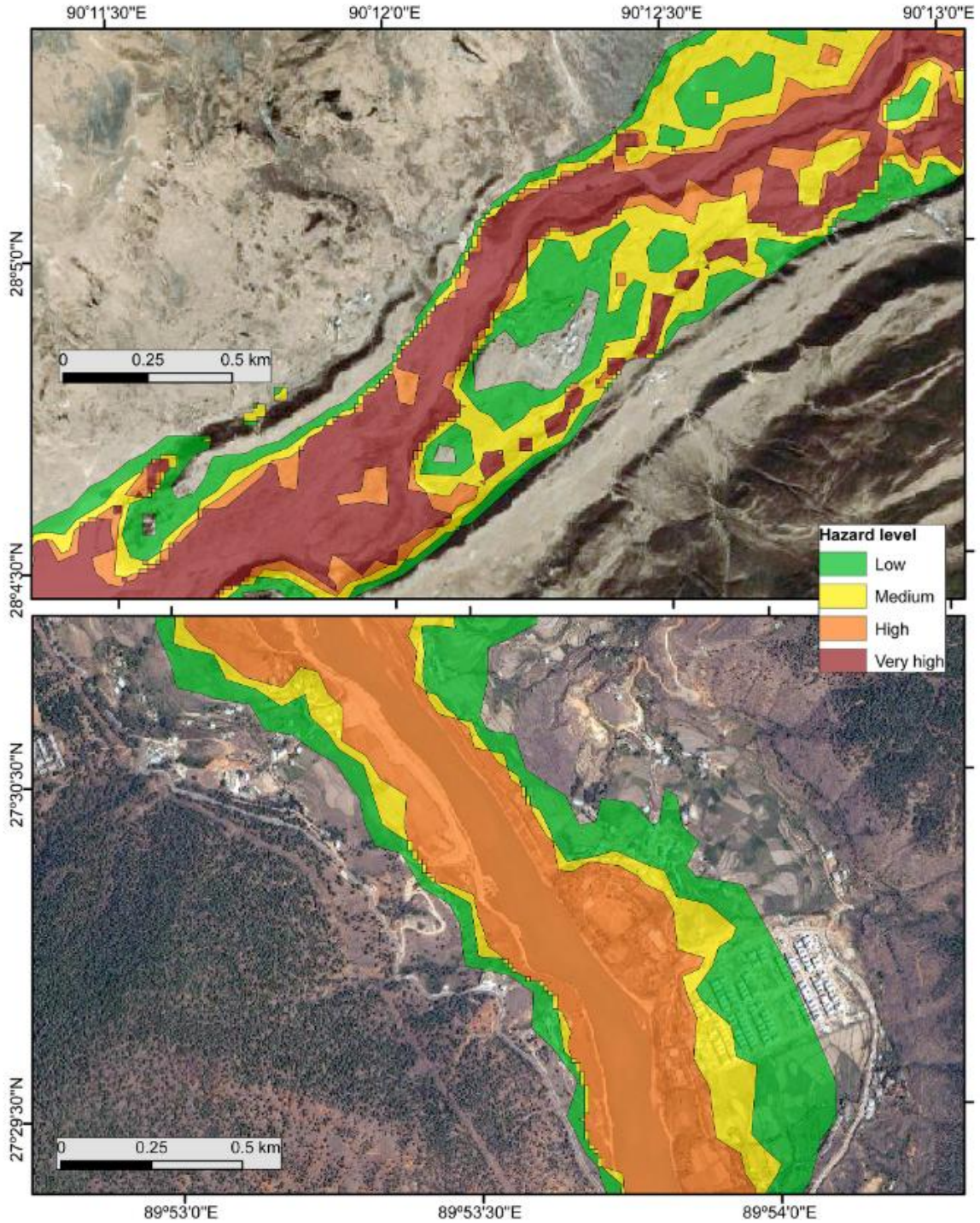


Fig. 3. Field photographs show the surrounding conditions that make lakes susceptible to GLOF, including calving front, avalanche, landslide/rockfall, and unstable moraine dam. Photo courtesy: NCHM, Bhutan.

Exposure


90°11'30"E
90°12'0"E
90°12'30"E
90°13'0"E
28°5'0"N
28°4'30"N
27°30'30"N
27°29'30"N
89°53'0"E
89°53'30"E
89°54'0"E
0 0.25 0.5 km
0 0.25 0.5 km
Hazard level
Low
Medium
High
Very high

Vulnerability

Table 2
List of indicators and corresponding represented components considered for calculating the social vulnerability index for the downstream settlements. The indicators are extracted from the Bhutan Population and Housing Census 2017 data (NSB, 2018), following the convention used by Allen et al. (2016) and Cutter et al. (2003). The absolute and normalized values for each indicator are also provided in supplementary Table S2.

No.	Social vulnerability indicator	Component represented
1	Old age dependence ratio	Sensitivity, capacity to prepare, respond and recover
2	Child dependence ratio	Sensitivity, capacity to prepare, respond and recover
3	Disability prevalence rate	Sensitivity, capacity to prepare, respond and recover
4	Female population	Sensitivity, capacity to prepare, respond and recover
5	Household without high school education attainment (adequate literacy)	Capacity to prepare, respond and recover
6	Household with insufficiency in food	Sensitivity, capacity to prepare, respond and recover
7	Rented household	Capacity to recover
8	Household without a reliable source of water	Capacity to prepare, respond and recover
9	Household without an improved sanitation	Capacity to recover
10	Household with more than a 30-minute walk to the road point	Capacity to prepare, respond and recover
11	Household without a smartphone	Capacity to prepare and respond
12	Household living in the house without housing concrete wall	Capacity to recover
13	Household without a car	Capacity to prepare, respond and recover
14	Household without a TV	Capacity to prepare and respond
15	Household without electricity as a main source of energy	Capacity to prepare and respond

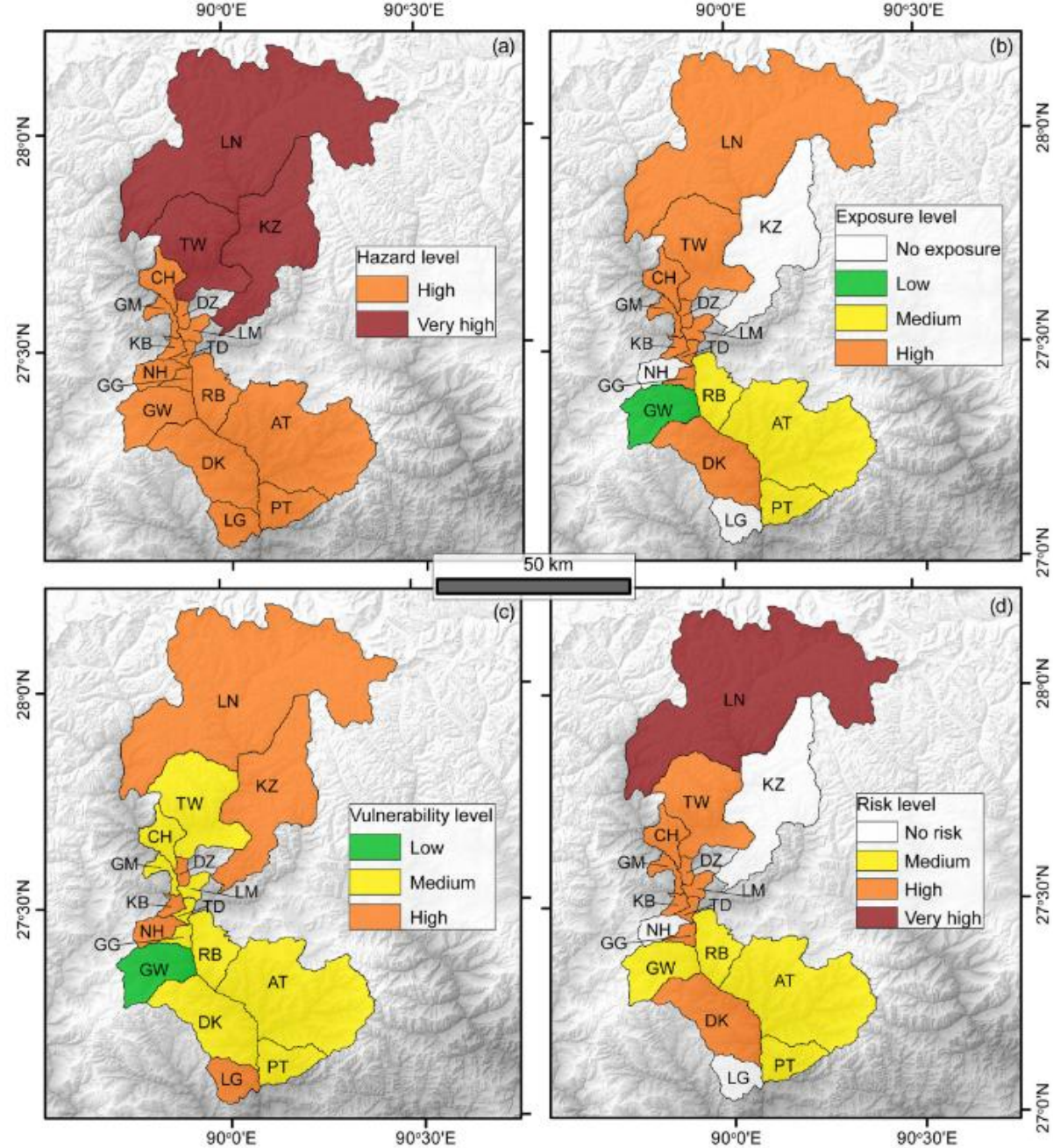


Fig. 9. The hazard (a), exposure (b), vulnerability (c), and risk (d) levels in 2020 of 17 subdistrict blocks downstream of the Lunana glacier complex under three different scenarios of GLOF magnitude (large, medium, and small) from each lake. The risk is calculated as a function of hazard, exposure, and vulnerability. The 17 subdistrict blocks are: Lunana (LN), Kazhi (KZ), Toedwang (TD), Chhubu (CH), Guma (GM), Dzomi (DZ), Lingmukha (LM), Kabisa (KB), Thedtsho (TD), Nahi (NH), Gase Tshogongm (GG), Gase Tshowongm (GW), Rubisa (RB), Darkar (DK), Athang (AT), Patshaling (PT), Largyab (LG). Background image: ALOS-PALSAR DEM.

RISK ASSESSMENT AND RISK MANAGEMENT PLAN

Risk Description	Risk Assessment	Mitigation Measures or Risk Management Plan
Construction and completion	Medium	The completion risk of any hydropower projects falls mainly on geological conditions and their relevant time and cost overruns. Many geological investigations and drilling studies undertaken for more than 3.5 years during project preparation found 80% of the rock tested to be in good order. The detailed project report prepared by a qualified engineering firm and the PPTA conducted due diligence studies including extensive field and laboratory investigations, such as hydrological, geological, geotechnical and geophysical, sedimentation, mechanical, and electrical requirements. Uncertainties of geological conditions remain as residual risks of construction delays and cost overruns, which could be affected by price increases in labor and materials. To address these risks, contingencies total $32 million or 22% of the base cost. Contractual mitigations are included in engineering, procurement, and construction contracts with liquidity damage clauses on a milestone basis. DGPC has adequate experience in procuring and managing contractors for construction of hydropower projects, demonstrated with Dagachhu hydropower plant implementation. Experienced staff of Dagachhu will support project construction. Project site accessibility is very good, i.e., along the main east–west national highway close to access roads and transmission connections.
Operations	Low	The plant's run-of-river hydropower operations are straightforward. DGPC has operated five run-of-river hydropower plants (HPPs) with capacity of 60 MW, 64 MW, 126 MW, 336 MW, and 1,020 MW. THyE will be staffed with experienced staff from DGPC and its subsidiaries—management and staff with adequate experience in operating HPPs. DGPC plans to select a reputed private joint venture partner with extensive experience in similar project development. The project's powerhouse will release water discharge to the Mangdechhu HPP reservoir (720 MW, run-of-river), expected to be available in 2017. The reservoir, under construction, is large enough to absorb the project's water release (confirmed by the Mangdechhu HPP's dam safety assessment under the PPTA due diligence studies). Project operation can begin regardless of completion of Mangdechhu HPP construction, as the existing Mangdechhu's diversion tunnel could easily absorb the project's water discharge.
Offtake	Low	A long-term 25-year PPA was agreed with PTC, the leading power trading company in India. DGPC has already exported almost 75% of its generation through other PPAs with PTC. The project's PPA is a take-or-pay contract in fixed volume terms for 80% of salable power; the volume will be less than 1% of the entire PTC trading power, which PTC can easily absorb. Given the prolonged power shortage in India, all of the project's power will be readily absorbed. If PTC fails to pay as per the PPA, the right for third-party sale will be triggered, but PTC's financial
Transmission	Low	The 132 kV transmission line works are contracted with BPC for project's full annual operations in 2020. The nearest substation u construction is on track to be available in 2017 as part of the ong Mangdechhu HPP. Once the project's transmission line (18.6 km connected to the substation, the project's power can alternatively exported through existing grid connections to India. The project's po will be mostly evacuated to India through the Mangdechhu–Gol Jigmeling–Alipurdwar transmission line corridor (400 kV) u construction for Mangdechhu operations scheduled in 2017.
Hydrological	Low	The hydrological risk primarily comprises drought risks, which significantly affect hydropower generation. The project's design energ based on 22-year hydrological data; the risk of any deviatio considered minimal. The financial analysis is based on the design en of 90% dependability (P-90 generation and plant availability of 9 which is quite conservative. This means that more water is likely t available for 90% of the operating time than designed on the hydrolo data. The assessment of potential future hydrological impacts, u various climate models found the project to be hydrologically soun projects that annual water flow in the Nikachhu watershed would incre by an average by 1%–3% for non-monsoon months and 5%–8% monsoon months to 2041–2060.
Other climate change (disaster)	Low	The climate change modeling analysis was extended to risks other potential hydrological issues, including risks of possible flo earthquakes, landslides, and glacial lake outbursts. The enginee design and environmental management plan are adaptive to extrem severe conditions created by such events. The analysis concludes overall climate change is unlikely to impose negative effects on project and its operation. Details of the assessment are in the clim change impact assessment. Effective adaptation measures will adopted for the project's structures, facilities, and adjoining area address such potential disasters as floods, landslides, and earthqua The dam safety design was reviewed and continues to be monitore DGPC's dam safety committee and a panel of experts. The project include an early warning system and awareness programs for pu safety in spite of no human settlement along the river downstream to confluence with the main river of the Mangdechhu.
Funding and debt	Low	The cofinancing loan from Indian commercial banks has been appro in principal. The cofinancing framework was agreed and its signing

Currency	Medium	Since the project is earned in Indian rupees from power exports, foreign debt from the OCR loan will have a currency exchange rate fluctuation

Risk Description	Risk Assessment	Mitigation Measures or Risk Management Plan
		risk if the Indian rupee depreciates. The Indian commercial bank cofinancing in local terms will reduce the overall debt service risk. The PPA stipulates that tariff adjustments in dollar terms may be introduced sometime in the future if both parties agree. Given the currency risk assessment, THyE will fully meet the debt service requirements even under an extreme scenario of continued annual 7% depreciation of Indian rupees over the debt repayment period. To mitigate the residual risks arising from the foreign debt services, THyE will adopt a foreign exchange fluctuation reserve fund. THyE and DGPC will also explore periodic currency hedging contracts once this becomes a feasible option in the financial market as covenanted.
Safeguards	Low	All safeguard documents are in accordance with ADB's Safeguard Policy Statement (2009). The compensation relating to resettlement will be fully paid in a satisfactory manner to the affected people. The current mitigation measures in the environmental management plan are appropriate and adequate to contain environmental impacts during project construction and operation, and environmental monitoring will be carried out during the project. To ensure minimum ecological flows, particularly in the lean seasons, the recommended level of 10% will continue to be monitored and finalized during implementation to ensure the habitat requirements combined with integrated water resources management, including relevant plant operations in the river basin. The ADB financing and loan agreements will ensure compliance with safeguard measures and implementation requirements, and an attached TA will support the safeguard activities. Extensive consultations and
		monitor and audit implementation of safe
Financial management	Low	DGPC has adopted comprehensive c covering accounting and financial procurement management, internal a safeguards, along with enterprise res external independent auditor reported t have adequate internal control system adequate competitive bidding, and s control to carry out their operations in a THyE will follow the same policies and pr
Procurement	Low	Increased transparency, efficiency, and ADB requirements for major constructior will conduct regular site inspections a timeliness of progress on contracts where
Corruption	Low	Increased transparency and accountab requirements and regular stakeholc agreements ensure that ADB's Antico measures are conformed with. ADB will related information on the website, and expenditure by ADB.
Overall	Low	

ADB = Asian Development Bank, BPC = Bhutan Power Corporation, DGPC = D kilovolt, MW = megawatt, OCR = ordinary capital resources, PPA = power pu preparatory technical assistance, PTC = Power Trading Corporation of India, Tangsibji Hydro Energy.
Source: Asian Development Bank.

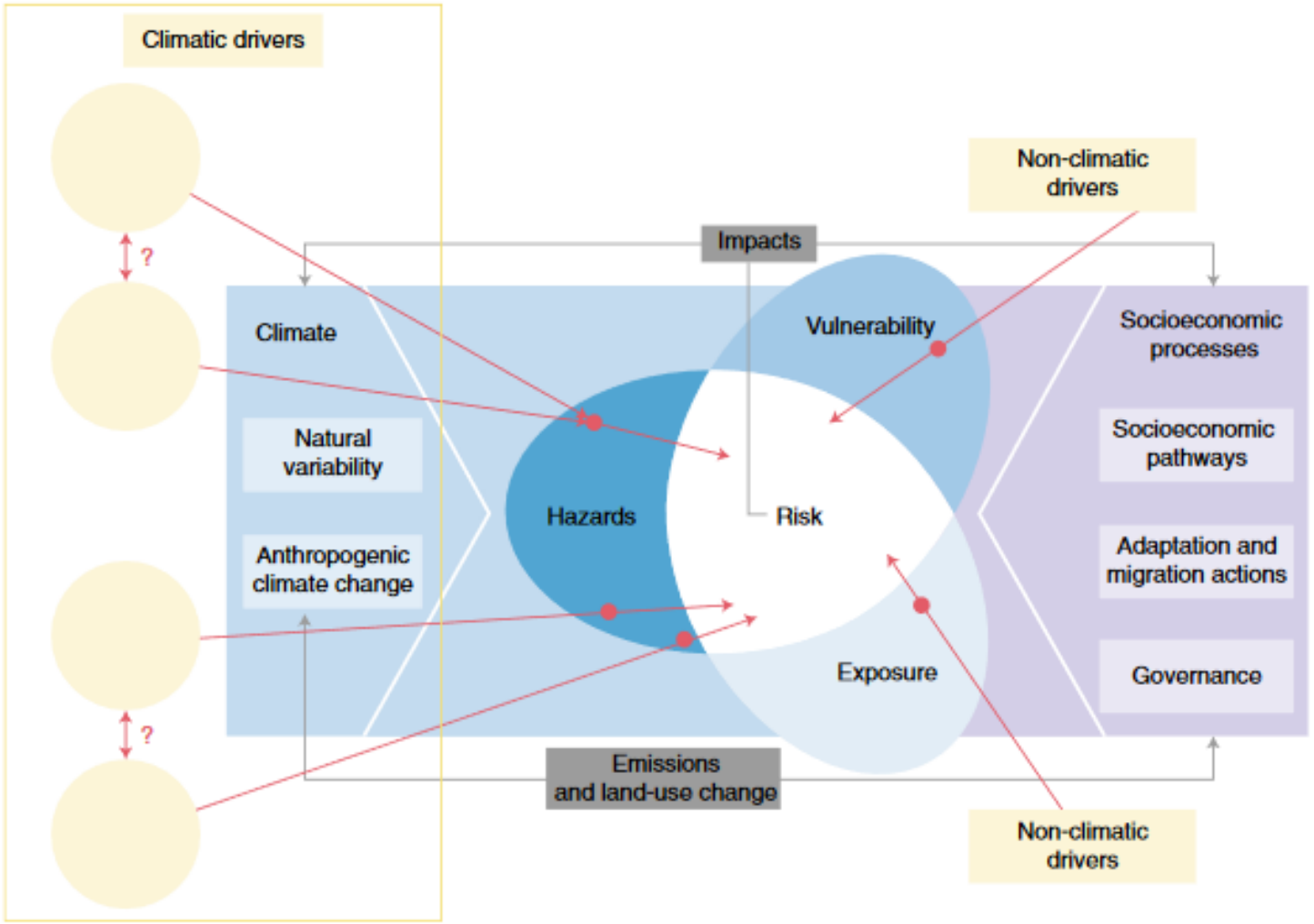
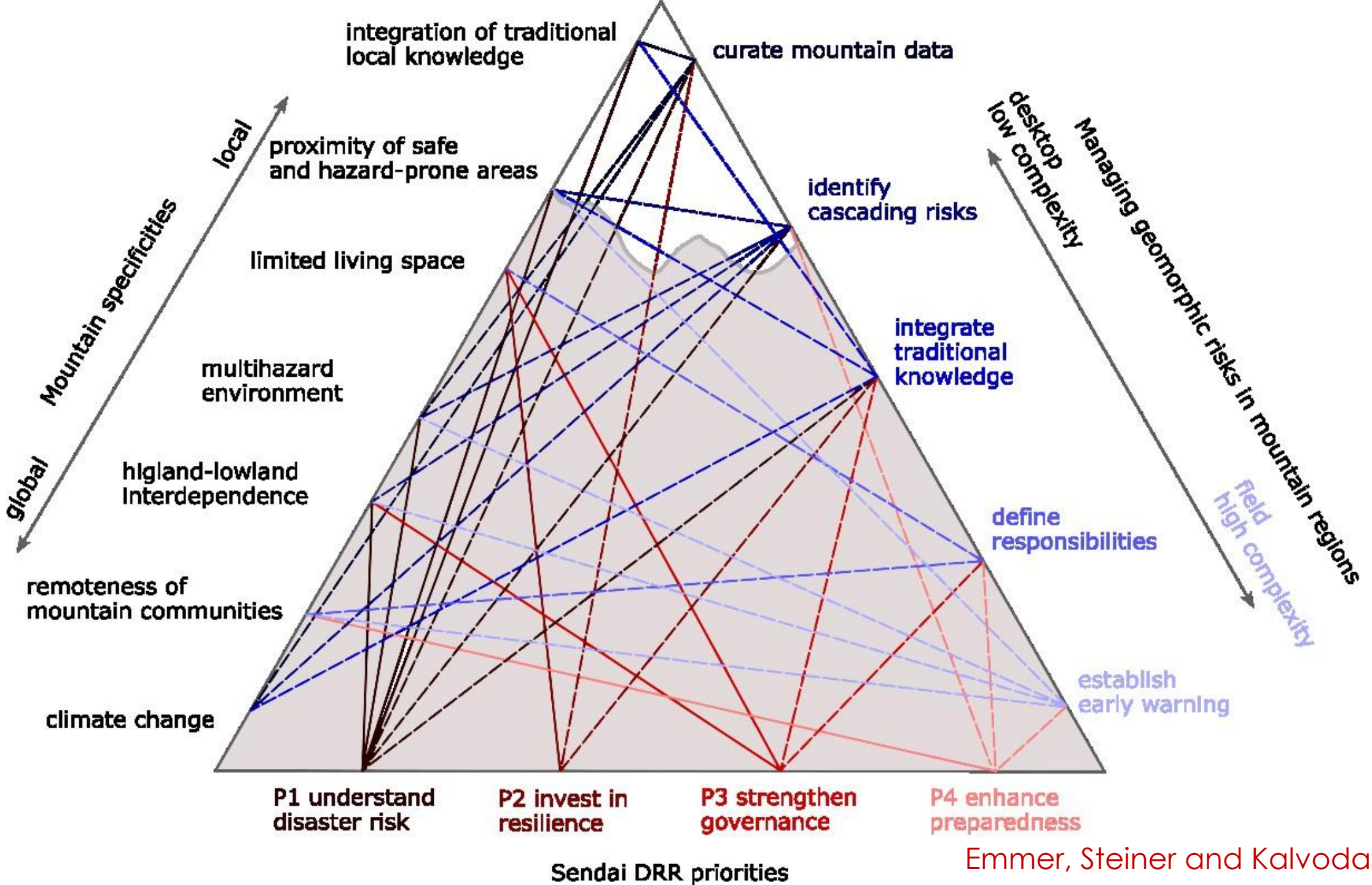


Fig. 1 | Extended risk framework. Multiple climatic drivers cause one or multiple hazards leading to societal and environmental risk. The climate drivers (which may vary from local-scale weather to large-scale climate modes, represented by yellow circles) and/or hazards may be mutually dependent. Non-climatic drivers related to vulnerability and exposure may also contribute to risk. Background risk figure adapted from ref. [96], IPCC.

Zscheischler et al. (2018)



Emmer, Steiner and Kalvoda (2024)

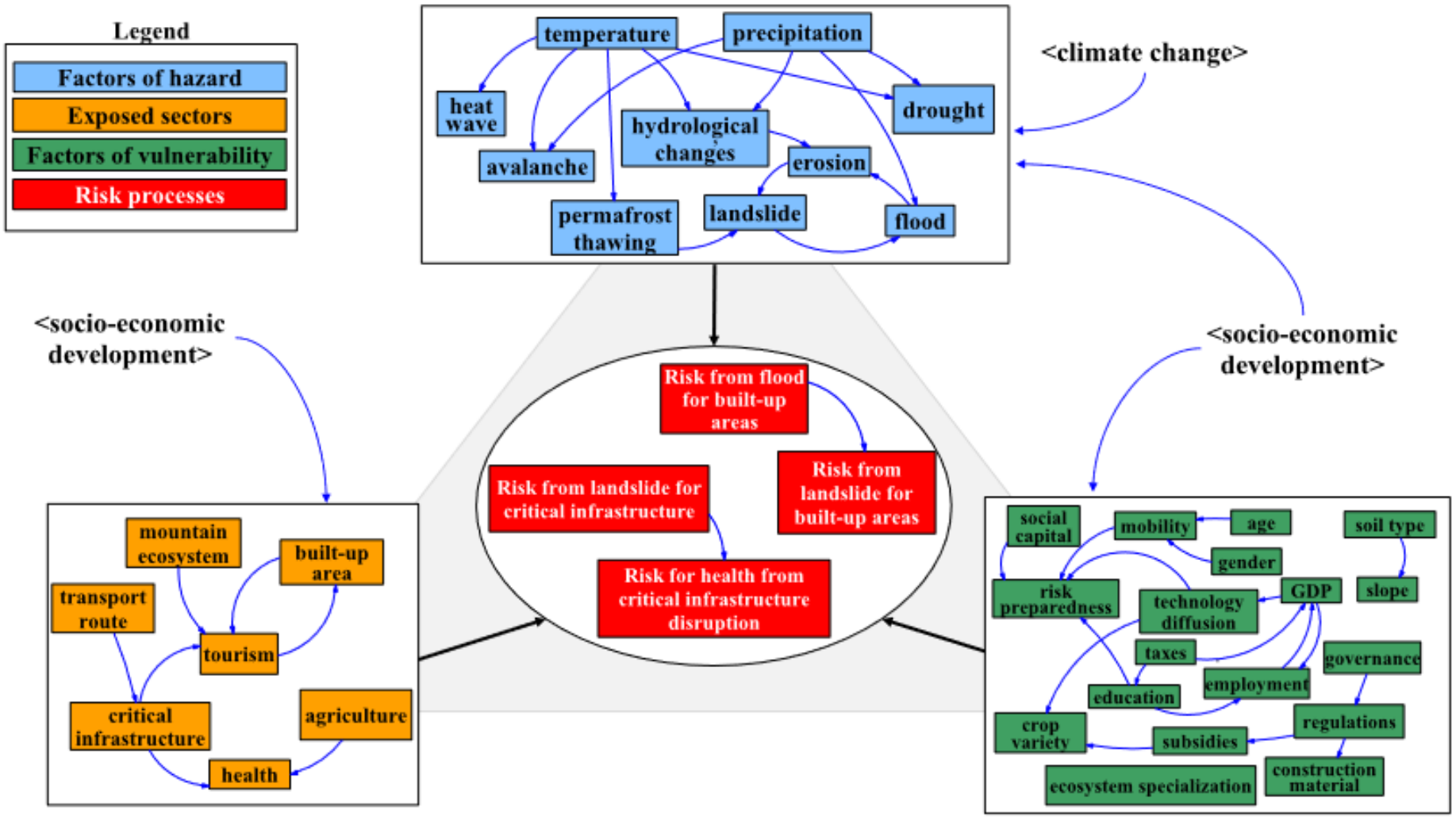


Fig. 2. Conceptual framework for mountain regions expanding the risk components into multi-hazard, exposure, multi-vulnerability and multi-risk.

Terzi et al. (2019)

Soneri Bank
بیرون ملک سے بھیجی جانے والی تمام رقوم سونیری بینک
کی تمام برانچز سے وصول کی جا سکتی ہیں
IME
International Money Express
Fast, reliable and secure remittance
MoneyGram.
bringing you closer
TRANSFAST
Worldwide Money Transfer
ALPINE EXCHANGE INC
sw small world
Money Transfer
GOLDEN
MONEY TRANSFER
WORLDWIDE
CASH EXPRESS
MONEY TRANSFER
الأنصاري للصرافة
AL ANSARI EXCHANGE
WALL STREET FINANCE WSF
ria Money Transfer

PIZZA PAMIR
TASTE OF MOUNTAIN HERBS
DINE IN / TAKEAWAY / DELIVERY
Pizza in the Mountains

LONG LIVE
PAK-CHINA FRIENDSHIP
中巴友谊万岁
PAK FC P52 OKZ
Pak-FC

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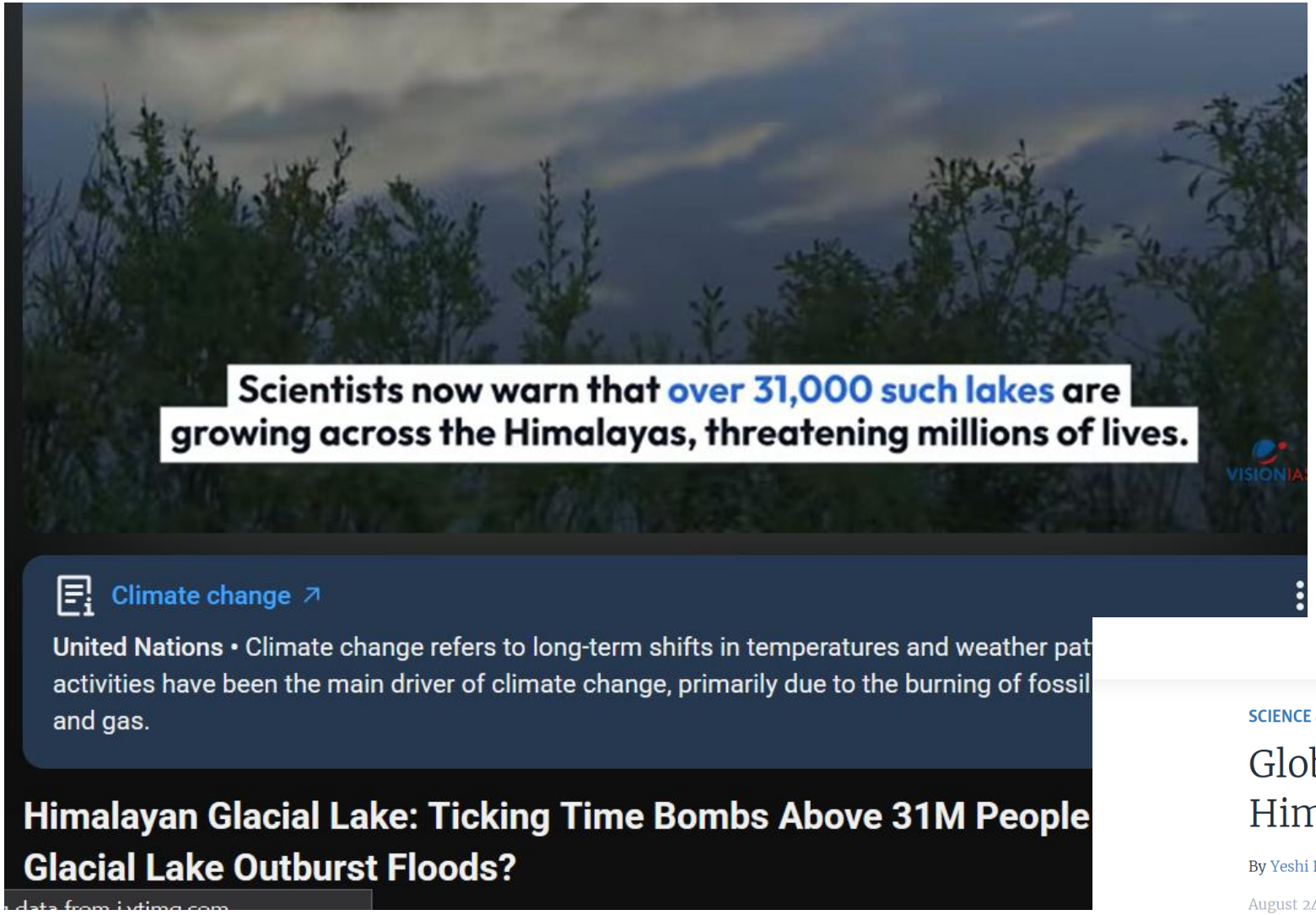
Concrete recent examples

Hazards: open data and policy (Glacial lake outburst floods)

Exposure: Connectivity and critical infrastructure (bridges)

Vulnerability: Mental health and mountain hazards

Hazards - Glacial lake outburst floods



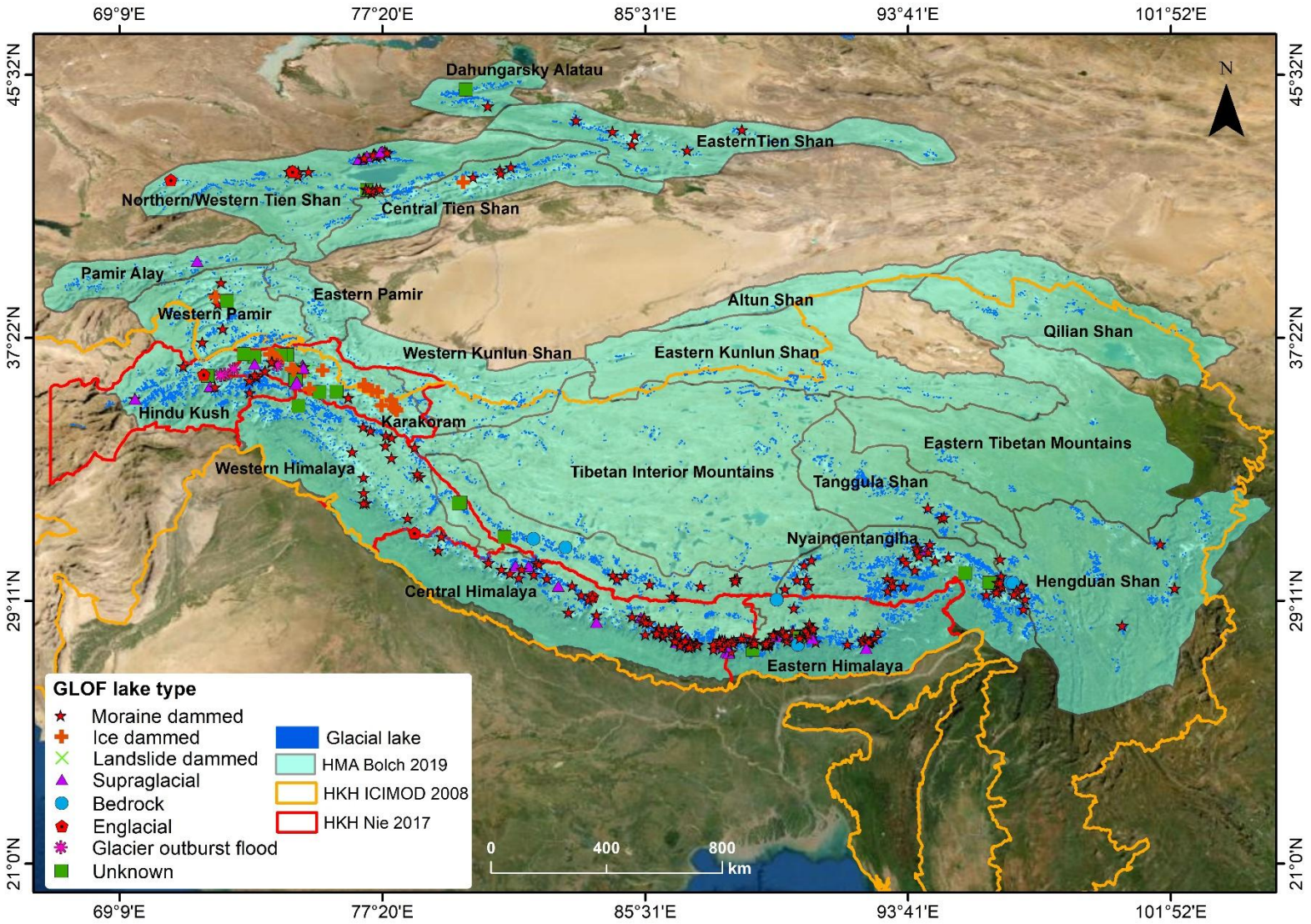
SCIENCE & HEALTH

Global Warming Increases Threat of Himalayas' Killer Lakes

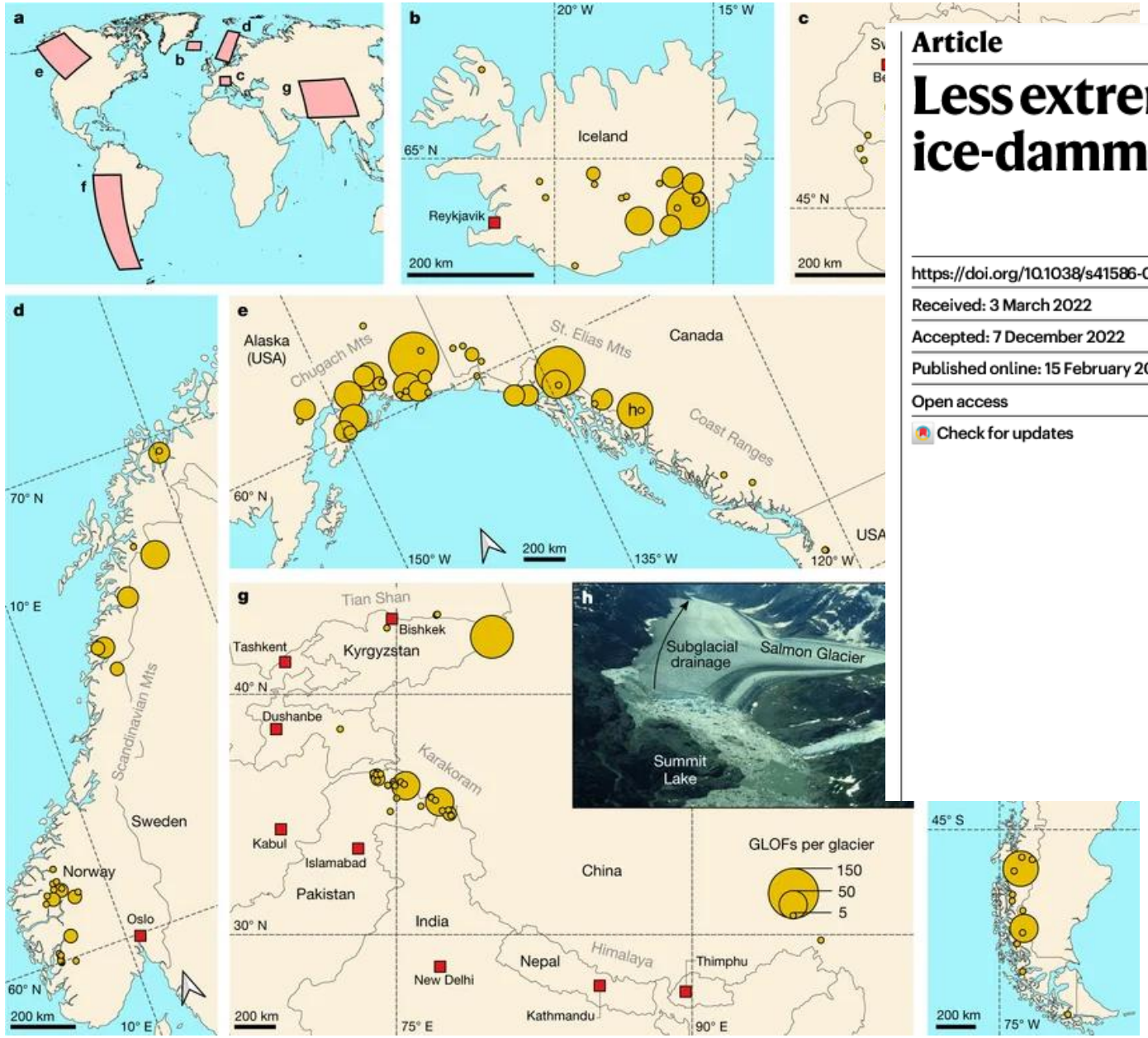
By Yeshi Dorje

August 24, 2019 8:22 PM

Glacial lake outburst floods



Shrestha et al. (2023)



Article

Less extreme and earlier outbursts of ice-dammed lakes since 1900

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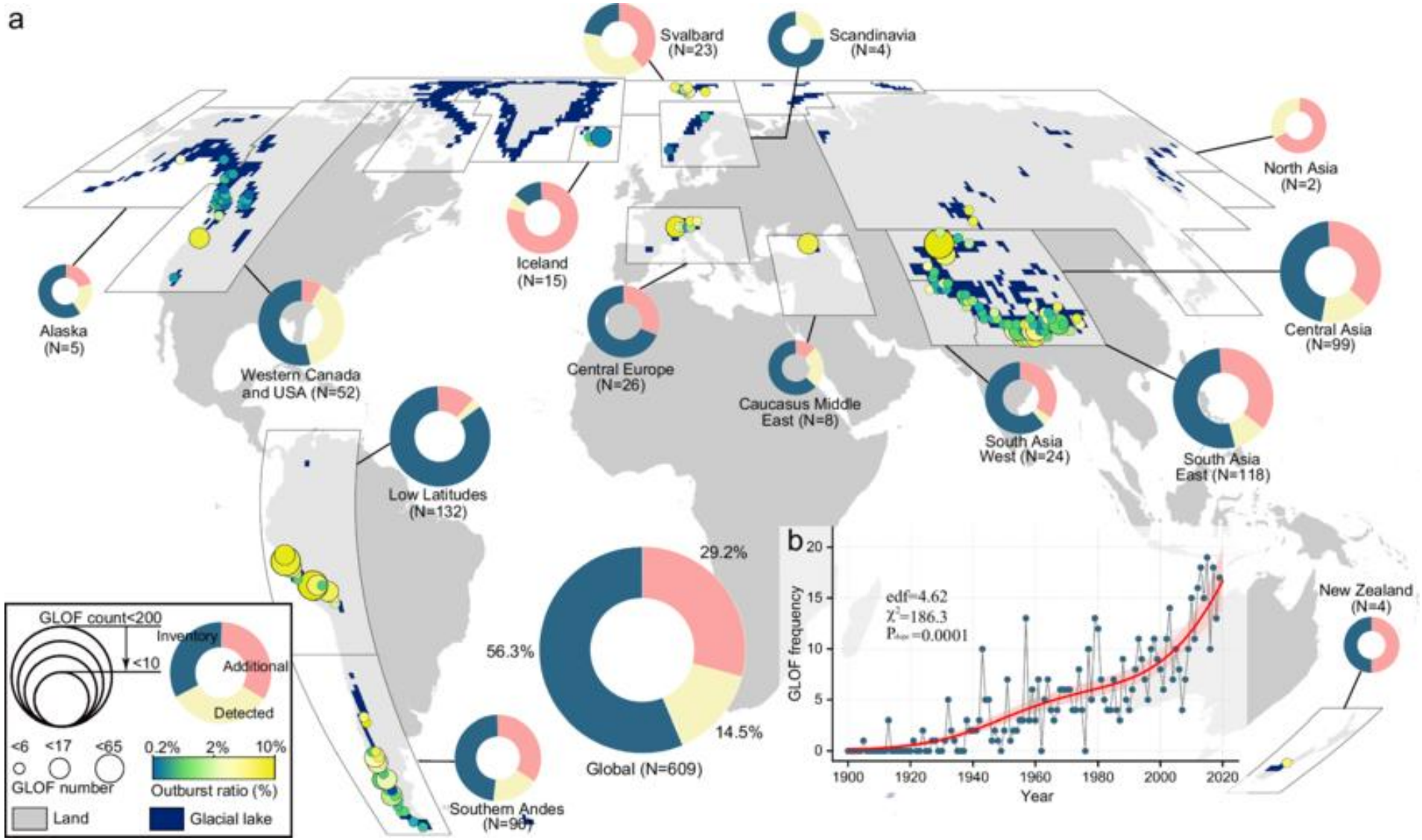
Open access

Check for updates

Georg Veh[1]✉, Natalie Lützow[1], Jenny Tamm[1], Lisa V. Luna[1,2,3], Romain Hugonnet[4,5,6], Kristin Vogel[7], Marten Geertsema[8], John J. Clague[9] & Oliver Korup[1,2]

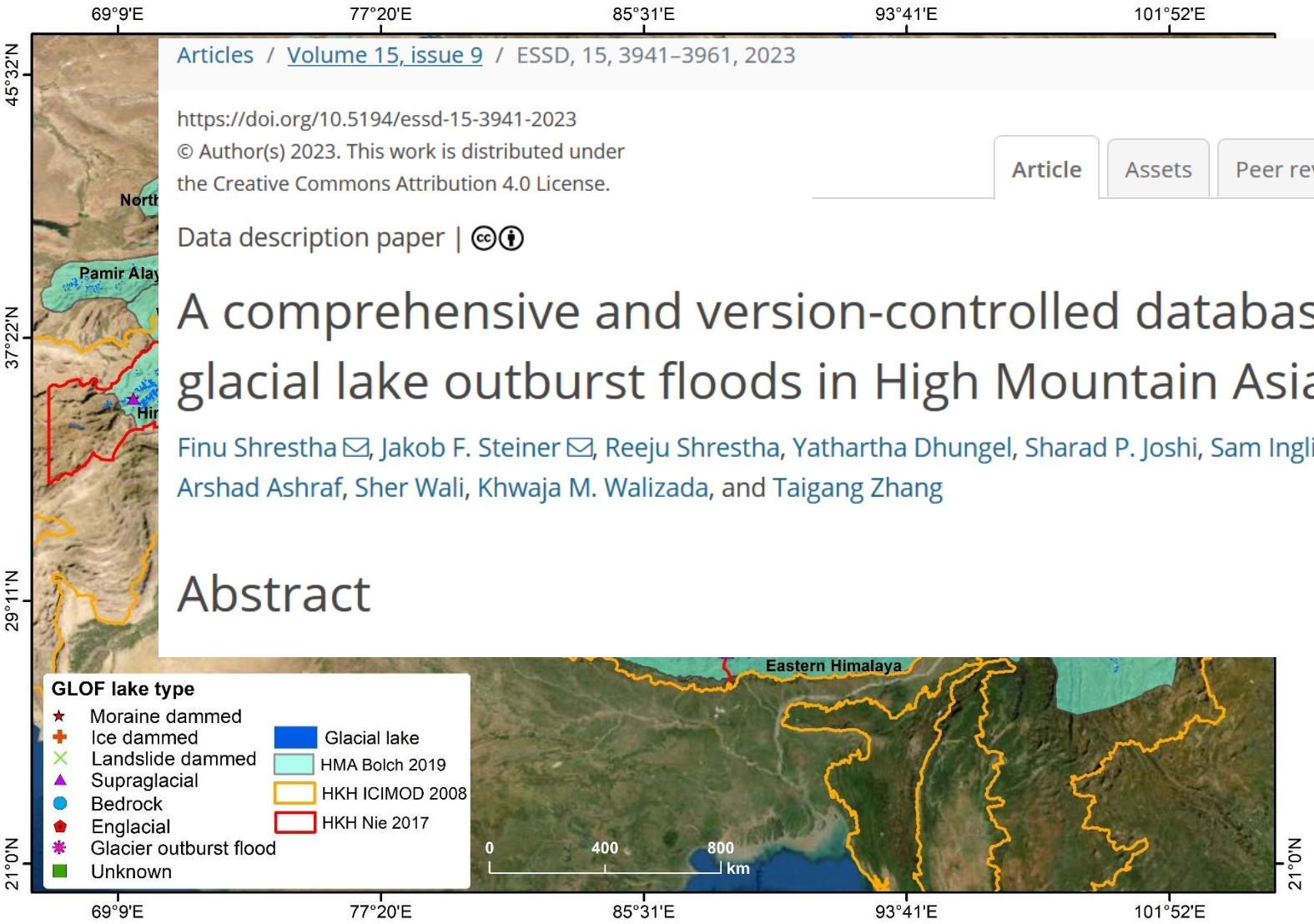
Episodic failures of ice-dammed lakes have produced some of the largest floods in history, with disastrous consequences for communities in high mountains[1–7]. Yet, estimating changes in the activity of ice-dam failures through time remains controversial because of inconsistent regional flood databases. Here, by collating 1,569 ice-dam failures in six major mountain regions, we systematically assess trends in peak discharge, volume, annual timing and source elevation between 1900 and 2021. We show that extreme peak flows and volumes (10 per cent highest) have declined by about an order of magnitude over this period in five of the six regions, whereas median flood discharges have fallen less or have remained unchanged. Ice-dam floods worldwide today originate at higher elevations and happen about six weeks earlier in the year than in 1900. Individual ice-dammed lakes with repeated outbursts show similar negative trends in magnitude and earlier occurrence, although with only moderate correlation to glacier thinning[8]. We anticipate that ice dams will continue to fail in the near future, even as glaciers thin and recede. Yet widespread deglaciation, projected for nearly all regions by the end of the twenty-first century[9], may bring most outburst activity to a halt.

Veh et al. (2023)



Zhang et al. (2025)

Glacial lake outburst floods



Articles / Volume 15, issue 9 / ESSD, 15, 3941–3961, 2023

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https://doi.org/10.5194/essd-15-3941-2023

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Article | Assets | Peer review | Metrics | Related articles

Data description paper |

05 Sep 2023

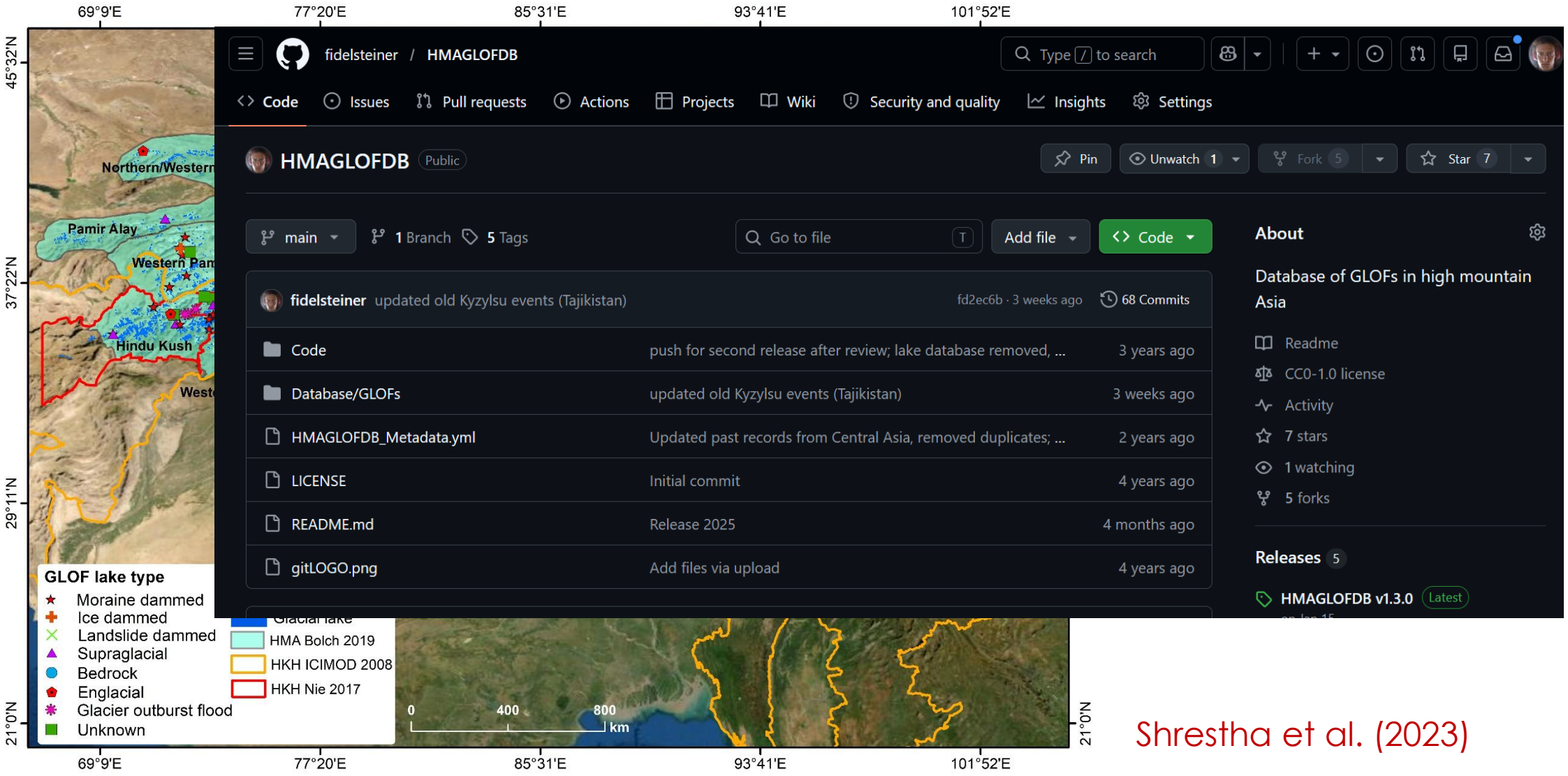
A comprehensive and version-controlled database of glacial lake outburst floods in High Mountain Asia

Finu Shrestha, Jakob F. Steiner, Reeju Shrestha, Yathartha Dhungel, Sharad P. Joshi, Sam Inglis, Arshad Ashraf, Sher Wali, Khwaja M. Walizada, and Taigang Zhang

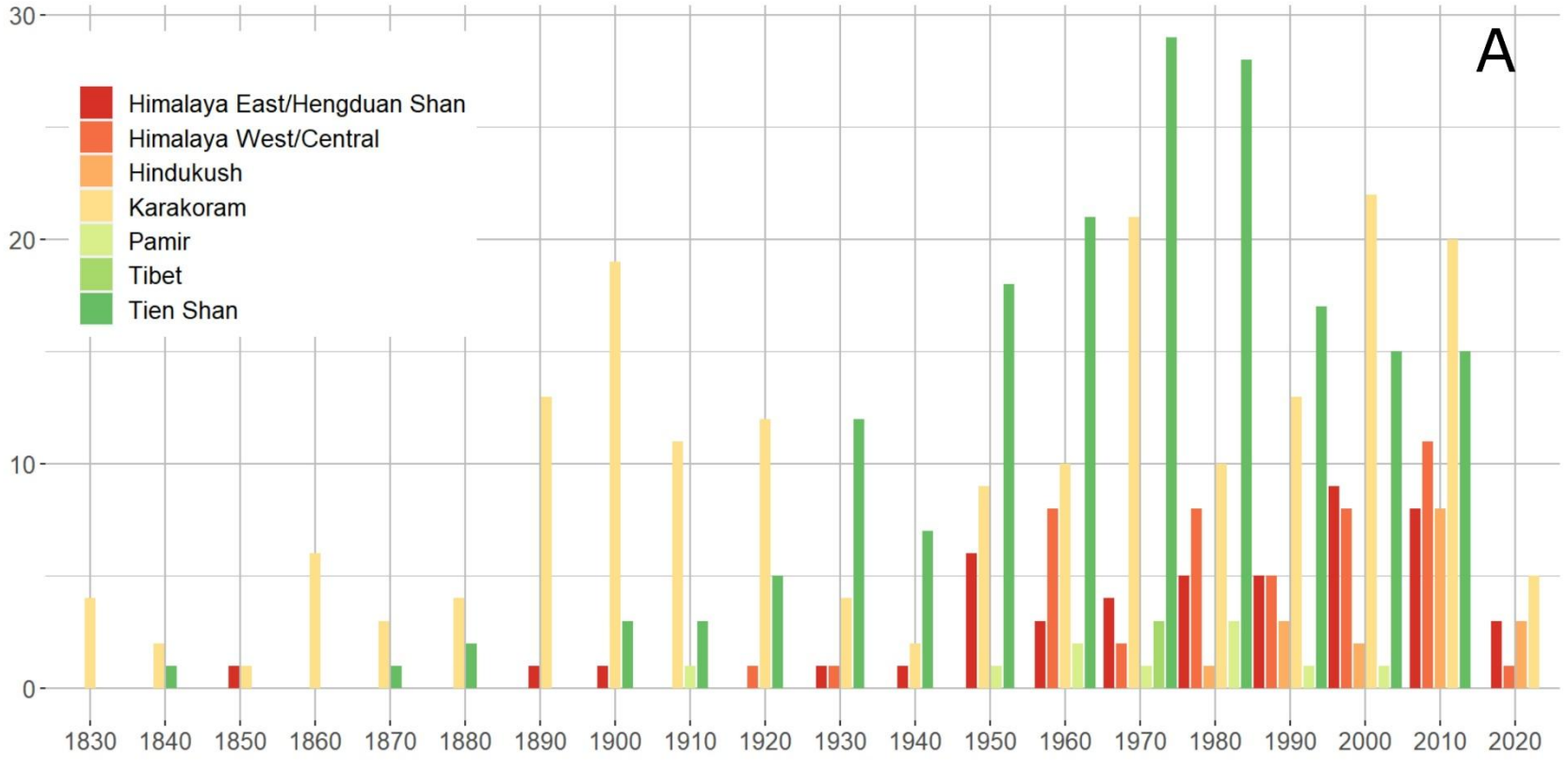
Abstract

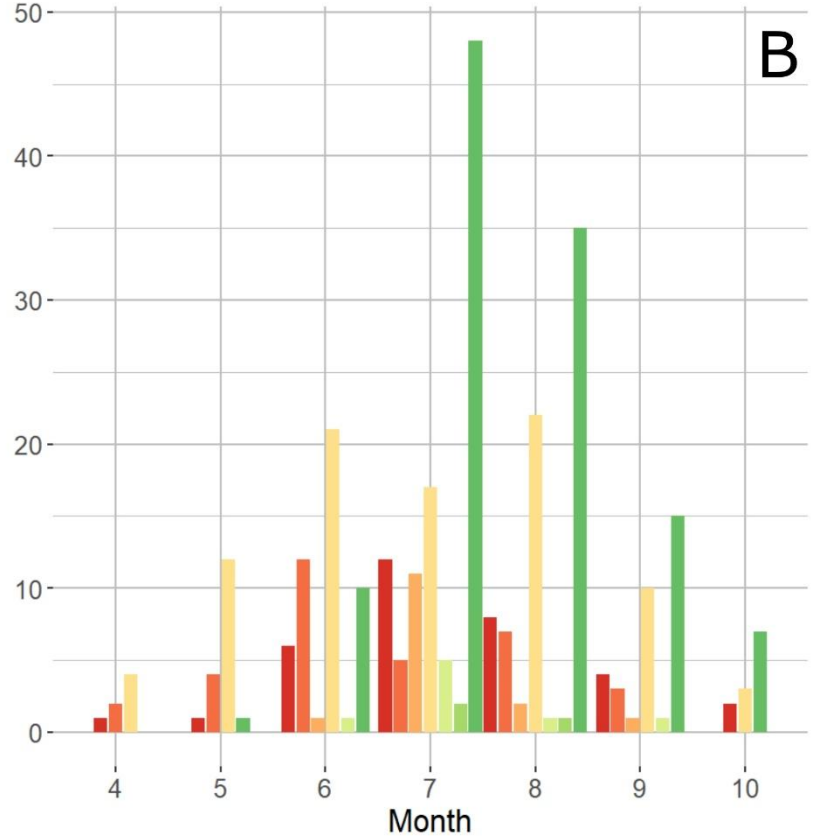
Shrestha et al. (2023)

Glacial lake outburst floods



Shrestha et al. (2023)

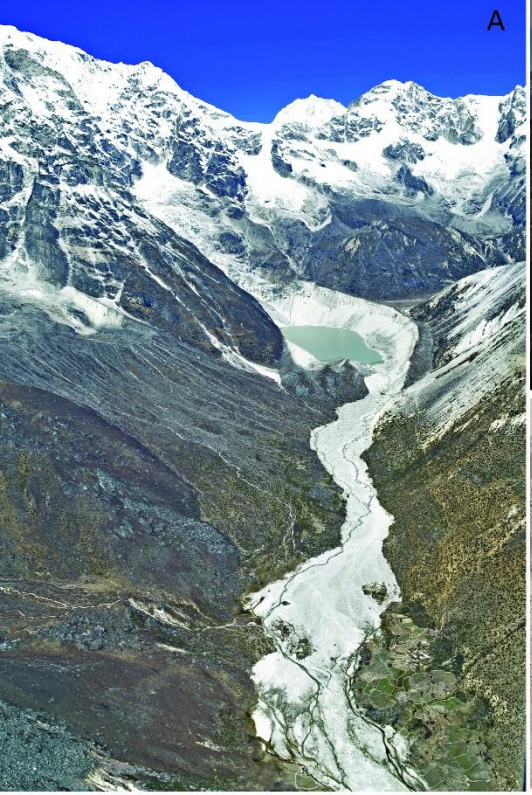


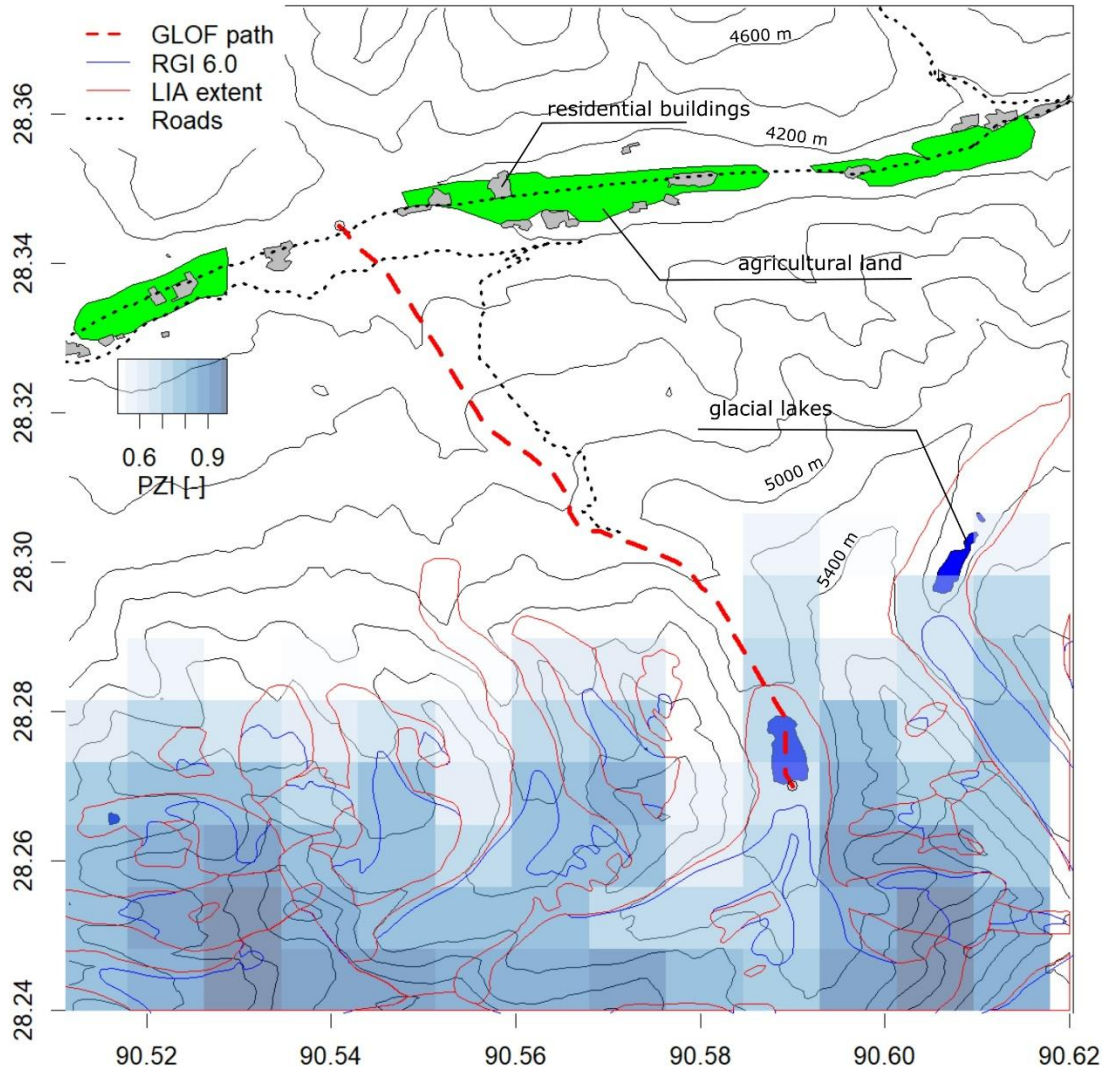


comprehensive inventory of GLOFs in HMA, including details on the time of their occurrence, processes of lake formation and drainage involved, and downstream impacts. We document 697 individual GLOFs that occurred between 1833 and 2022. Of these, 23 % were recurring events from just three ephemeral ice-dammed lakes. In combination, the documented events resulted in 690 fatalities of which 906 can be attributed to 24 individual GLOF events, which is 3 times higher than a previous assessment for the region. The integration of previous inventories of glaciers and lakes within this database will inform future assessments of

Shrestha et al. (2023)

Shrestha et al. (2023)





Shrestha et al. (2023)

Exposure - Bridges

Infrastruktur-Sanierung

Von wegen Brücken-Turbo

6. November 2025, 11:57 Uhr | Lesezeit: 2 Min. | 4 Kommentare

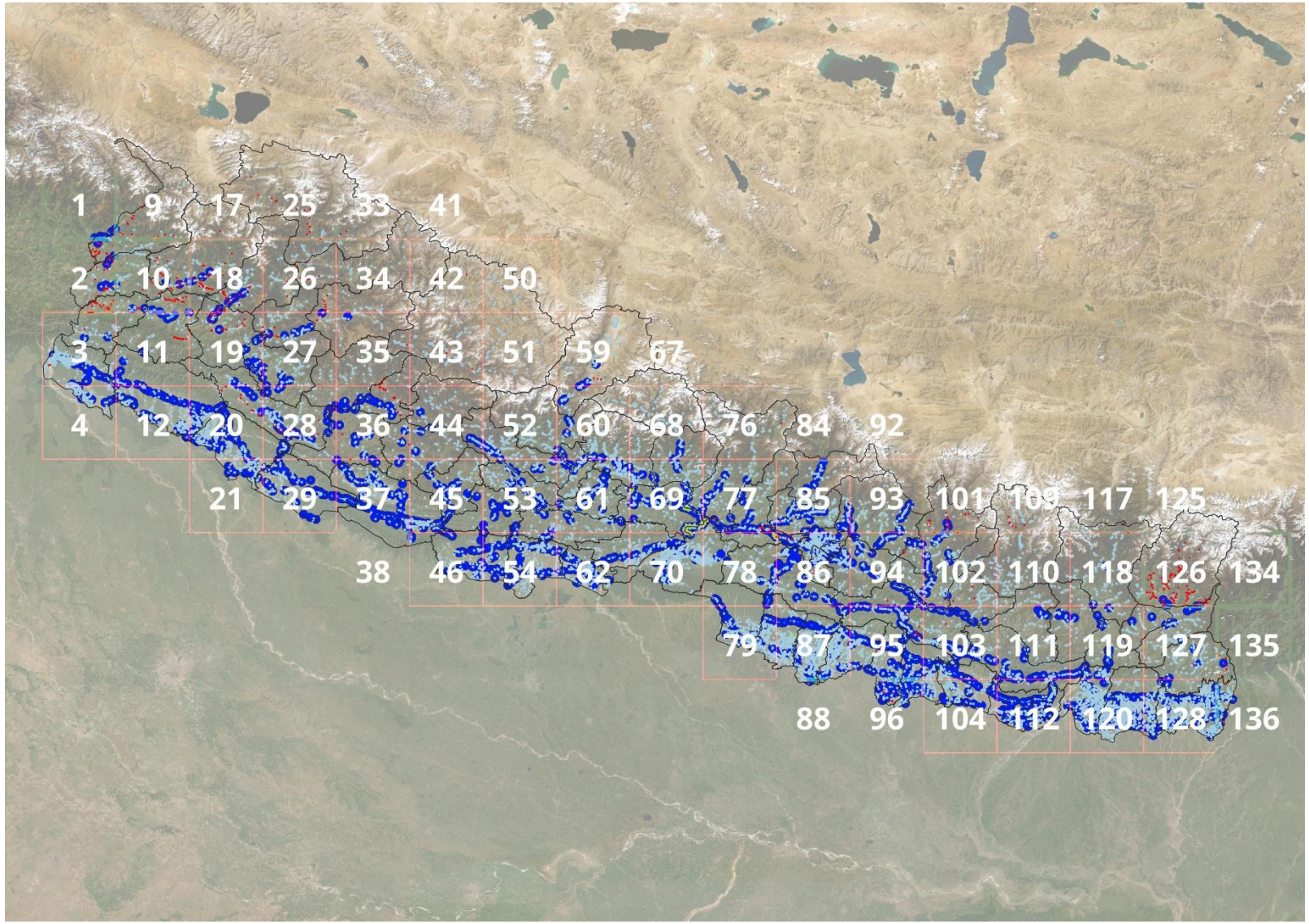
Schon im April hatte der Bundesrechnungshof das Tempo der Brückensanierungen in Deutschland kritisiert.
Bernd Weißbrod/dpa

KI Zusammenfassung

4000 Brückenbauwerke will die Regierung bis 2032 sanieren. In diesem Jahr hat das gerade mal bei 170 geklappt. Das könnte noch enorm teuer werden.

2000 bridges in Department of Roads database

20000+ bridges actually existing



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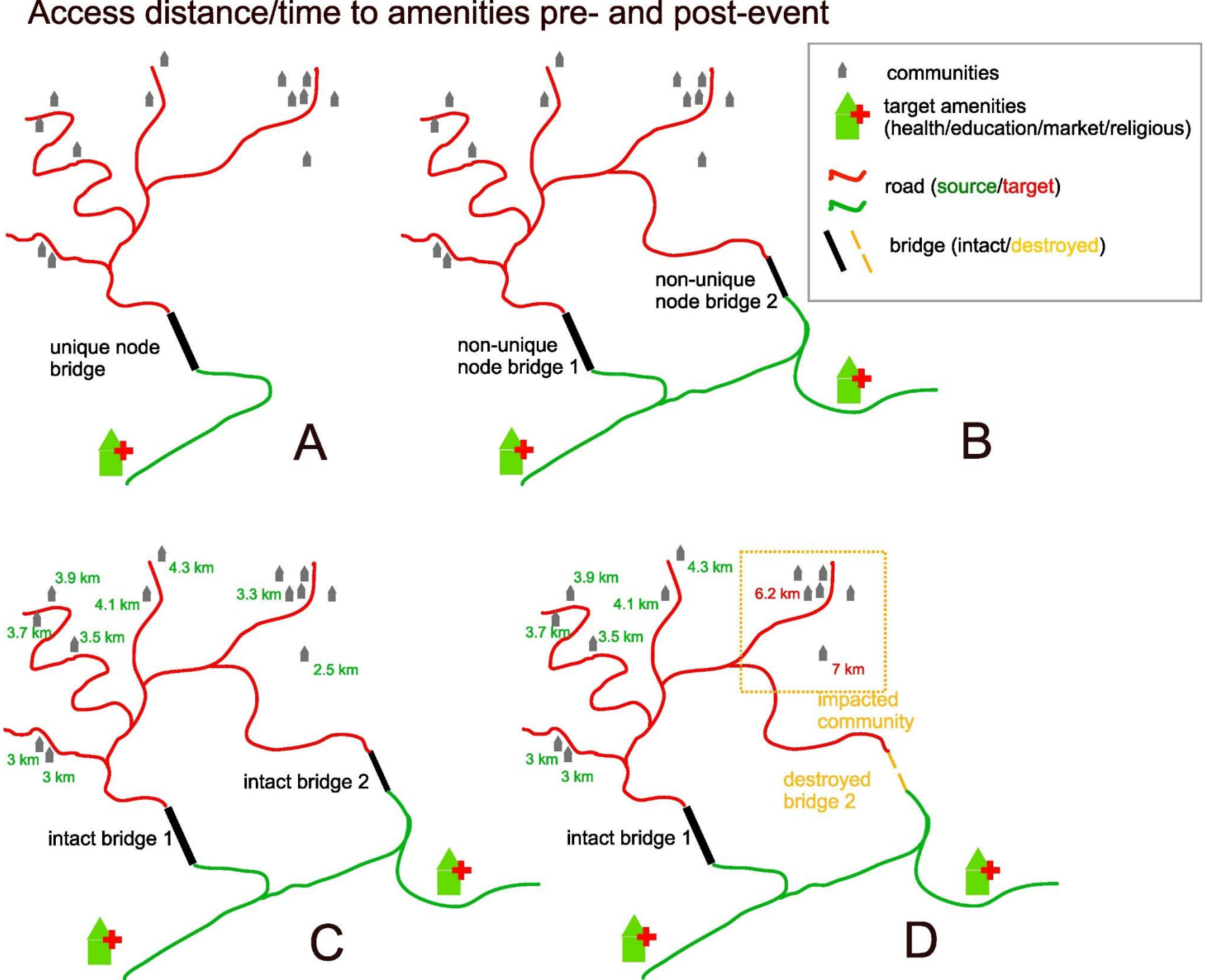
Access distance/time to amenities pre- and post-event
communities
target amenities
(health/education/market/religious)
road (source/target)
bridge (intact/destroyed)
unique node
bridge
A
non-unique
node bridge 1
non-unique
node bridge 2
B
3.9 km
4.3 km
4.1 km
3.3 km
3.7 km
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impacted
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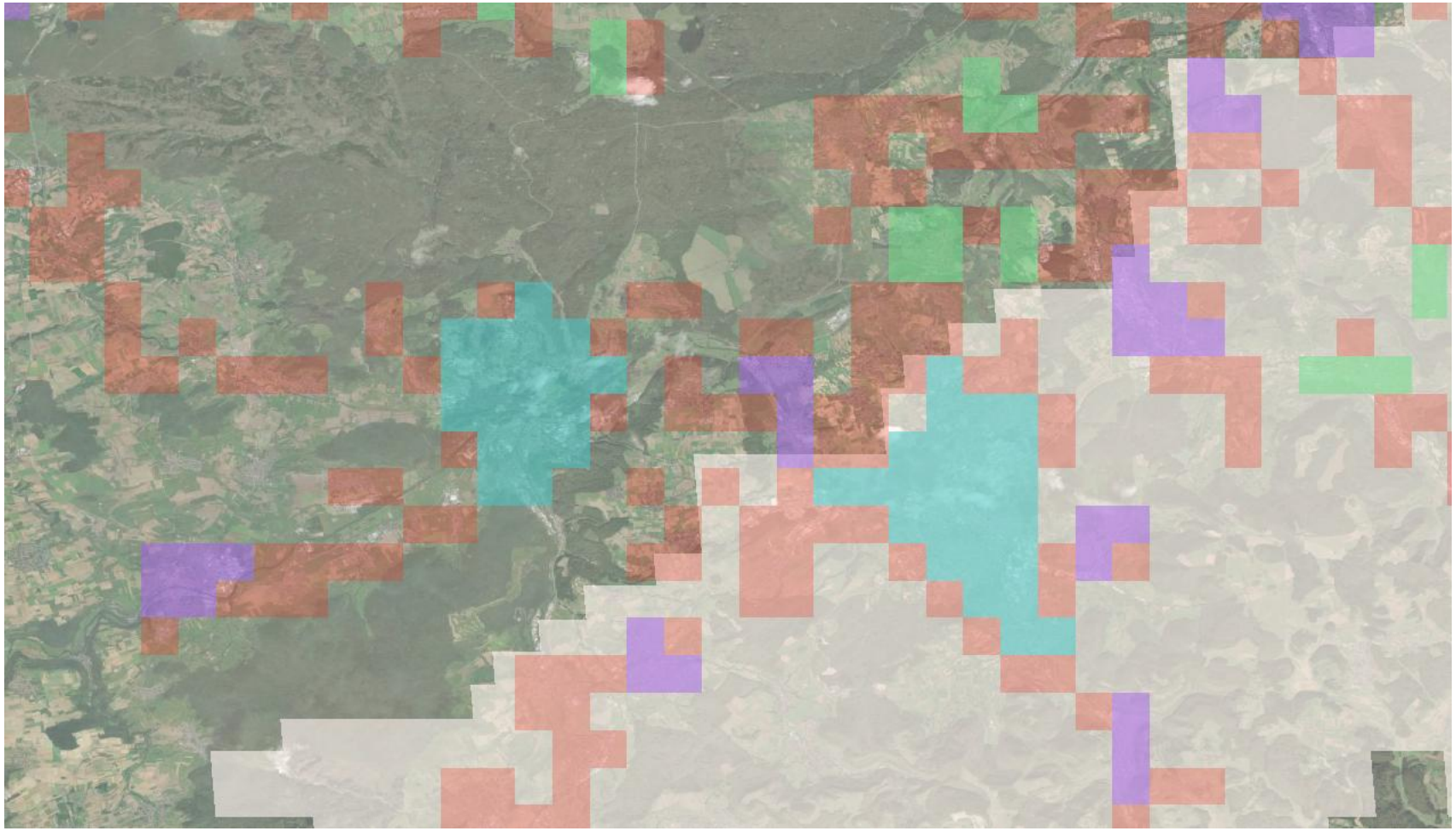
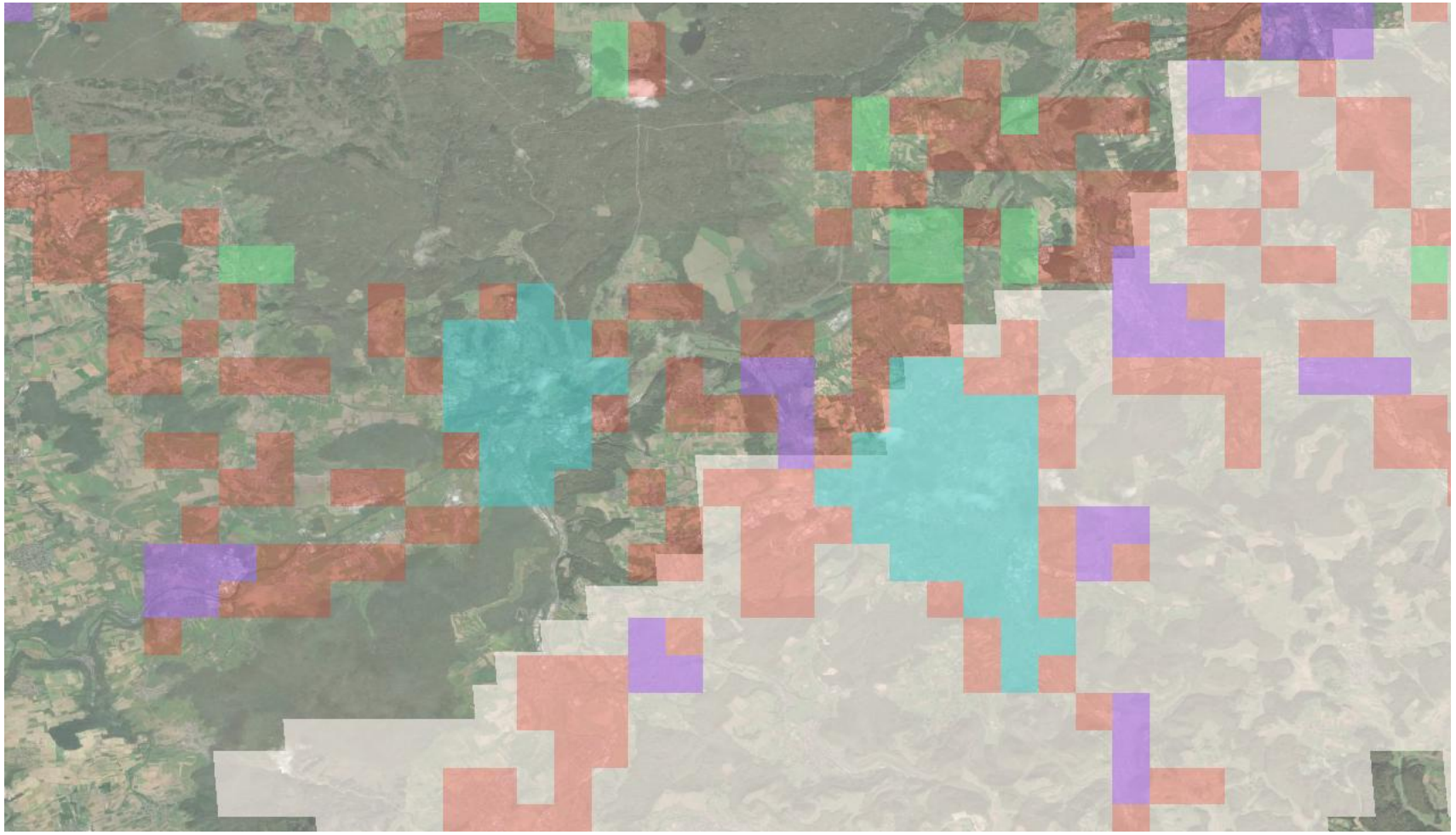
3 km
3 km
destroyed
bridge 2
intact bridge 1
D

Exposure – urban space

Tübingen 2010



Tübingen 2025



Vulnerability – mental health and mountain risks

Climate change and mental health: risks, impacts and priority actions

It is well understood that human health is threatened by the impacts of climate change [1,2,3]. In the 2017 Lancet Countdown on Climate Change and Health, authors state: "The human symptoms of climate change are unequivocal and potentially irreversible—affecting the health of populations around the world today" [4]. Climate change is no longer a looming threat but rather a destructive reality with dire predictions for the future. The World Health Organization (WHO) estimates an increase of 250,000 excess deaths per year between 2030 and 2050 due to the "well understood impacts of climate change" [5]. Impacts include heat-related morbidity and mortality, increases in vector-borne diseases (e.g. dengue fever, malaria), increased respiratory illness, and morbidity and mortality due to extreme weather events [6, 7]. The lesser-known, and often overlooked, effects of climate change include the risks and impacts to mental health—the focus of this article.

Hayes et al. (2018)

Vulnerability – mental health and mountain risks

Climate change and mental health: risks, impacts and priority actions

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ental health crisis

0

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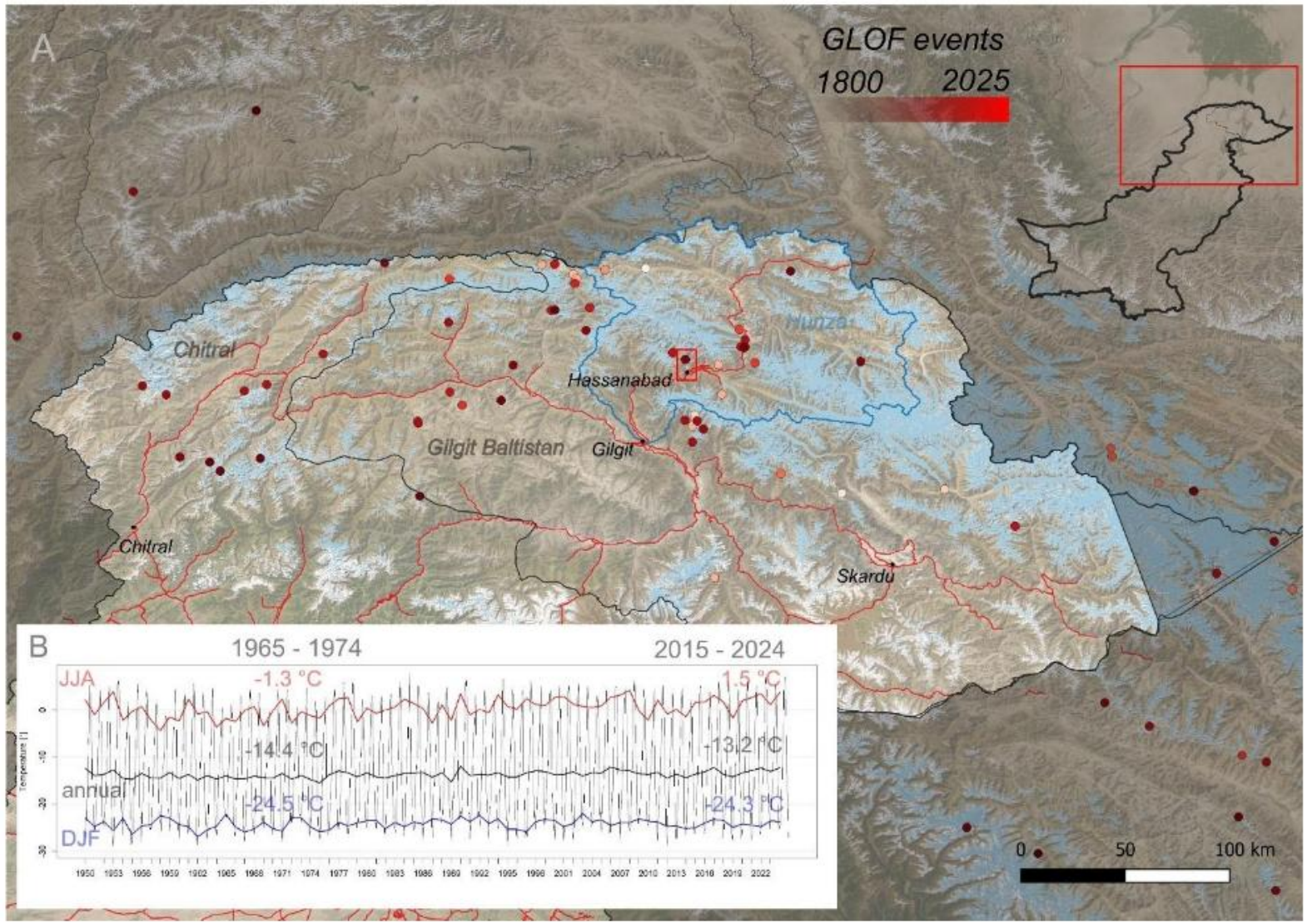
ADD **DAWN** AS A TRUSTED SOURCE Add as a preferred source on Google

Gilgit-Baltistan (GB), particularly in
ntal health crisis, prompting the
ical rehabilitation efforts in the
disaster-hit areas.

According to mental health professionals, a rising number of individuals from flood-affected regions have reported trauma-related psychological issues, highlighting the urgent



Aaliya et al. (in review)

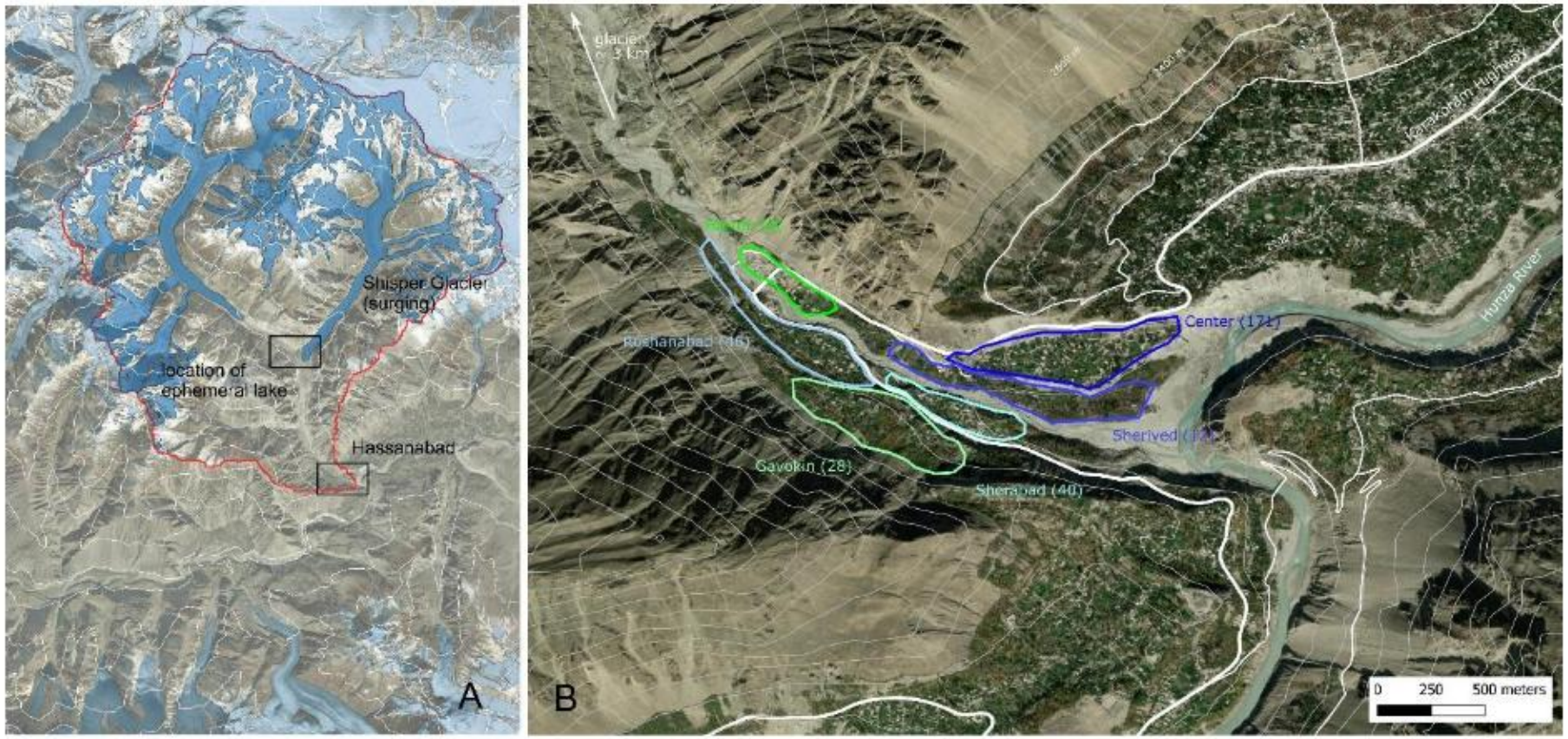


Fig. 2. Shisper Basin, Hassanabad catchment, and the location of the ice-dammed lake are shown in (A). B represents the six settlements of Hassanabad, with each number indicating the number of participants surveyed.

Aaliya et al. (in review)

Table 1. Summary of the sociodemographic information of the surveyed respondents.

Demographic variables	Features	Gavokin	Harum	Hassanabad Center	Roshanabad	Sherabad	Sherived
Age	(Mean± *SD*)	38± 13.1	35.2±12.4	37.1± 12.2	42±14	36± 13	41.4± 13.4
Gender	Male (%)	28.6	41.7	42.7	45.7	42.5	50
	Female (%)	71.4	58.3	57.3	54.3	57.5	50
Family Size	(Mean± *SD*)	8.2± 3.1	8.3± 2.4	8± 3.2	7.3± 3.1	8± 3.4	5.2±1.4
Employment Status	Employed (%)	25.0	17	42	35	40.0	50
	Unemployed (%)	75.0	83.3	59	65.2	60.0	50
Schooling	Schooling (%)	86	75	87.1	83	88	100
	No Schooling (%)	14.3	25	13	17.4	13	0.0
Monthly Income (PKR)	(Mean± *SD*)	10,179± 20971.0	10,333.3± 24373.4	17,175.4± 28181.0	40,348± 152558	18,125.0± *28027*	13,375.0 ±19705

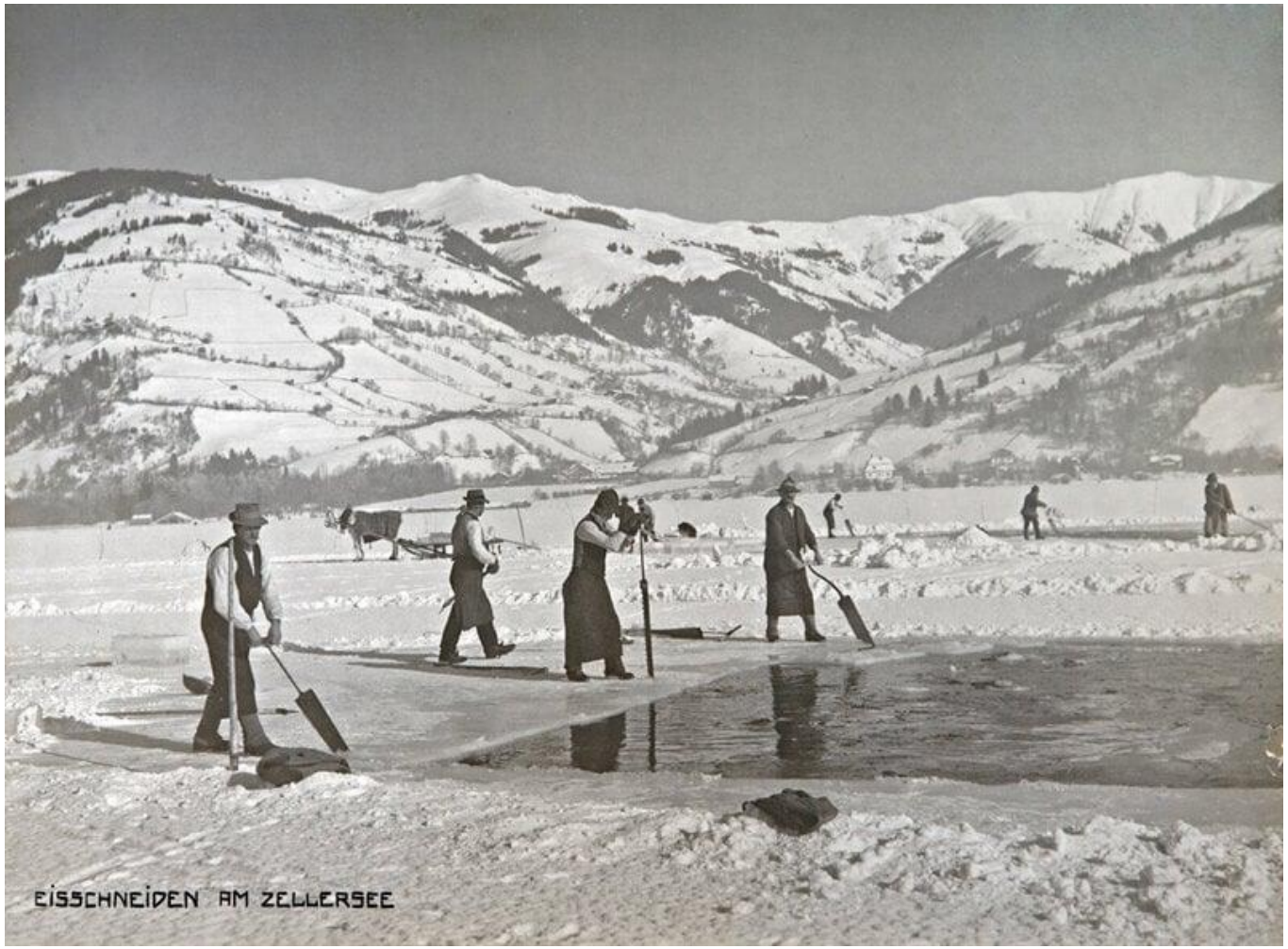
Aaliya et al. (in review)

Adaptation

Photo: Anna-Maria Walter

Photo: Zahir Ahmad

Video: Zahir Ahmad


EISSCHNEIDEN AM ZELLERSEE

Course content

28 May (0900 – 1500)

09:00 – 09:45	Disaster Risk Reduction
09:45 – 10:00	Break
10:00 – 10:45	Concrete recent examples
10:45 – 11:00	Break
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International initiatives addressing mountain risks

Loss and Damage (https://www.lossanddamagecollaboration.org)

Mountain Research Initiative (MRI) (https://mountainresearchinitiative.org/)

Adaptation at Altitude (https://adaptationataltitude.org/)

Mountain Partnership (https://www.fao.org/mountain-partnership/home/en/)

UNFCCC and Loss and Damage

United Nations Framework Convention on Climate Change (UNFCCC)

- An international treaty to limit dangerous climate change
- Signed by 154 states in 192 (Rio de Janeiro)
- Secretariat in Bonn, Germany
- Hosts the Green Climate Fund (GCF) – goal of $100 billion annually by 2020 – the Adaptation Fund (AF), as well as smaller LDCF, SCCF and GEF
- Invites the IPCC (Intergovernmental Panel on Climate Change) to provide assessments
- Conference of Parties (COPs) are arranged under the framework (many of them!)

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Home

UNCCD 16th session of the Conference of the Parties

RIYADH, SAUDI ARABIA | 2-13 DECEMBER 2024

COP 16 Overview | Registration | High-level segment | Official documents | Daily journal | Special and side events | Media

Loss and Damage Fund

[…] Parties recognised by the UNFCCC as the most responsible for the climate crisis would provide predictable, and new and additional finance to address loss and damage.

[…] Loss and Damage Fund will tackle the gaps that current climate finance institutions such as the Green Climate Fund (GCF) and humanitarian assistance […] do not fill on Loss and Damage. The current international climate finance system is already stretched to breaking point due to insufficient funds, lack of political will and poor quality climate finance that indebts countries.

Loss and Damage Fund

[…] Parties recognised by
provide predictable, and

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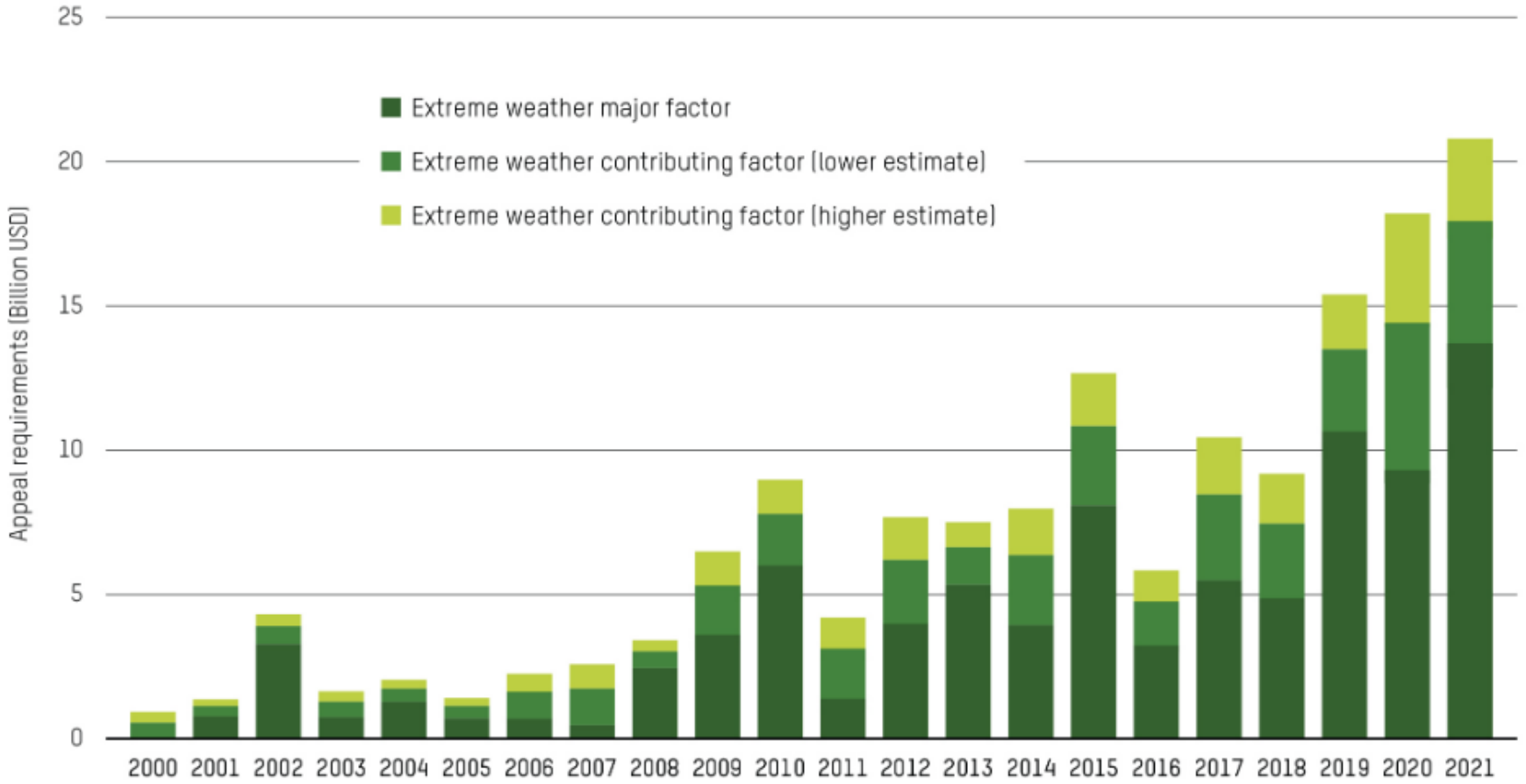
indebts countries.

Figure 3: Funding requirements for UN humanitarian appeals linked to extreme weather, 2000–2021



USD amounts are in 2021 real terms. The dark green sections cover appeals for which extreme weather was a major factor, and therefore 100% of the appeal requirement is counted. The higher estimate (lightest green) counts 50% and the lower estimate (middle green) counts 30% of the value of these appeals.

https://oxfamilibrary.openrepository.com/bitstream/handle/10546/621382/bp-fair-finance-loss-and-damage-070622-en.pdf

Loss and Damage Fund

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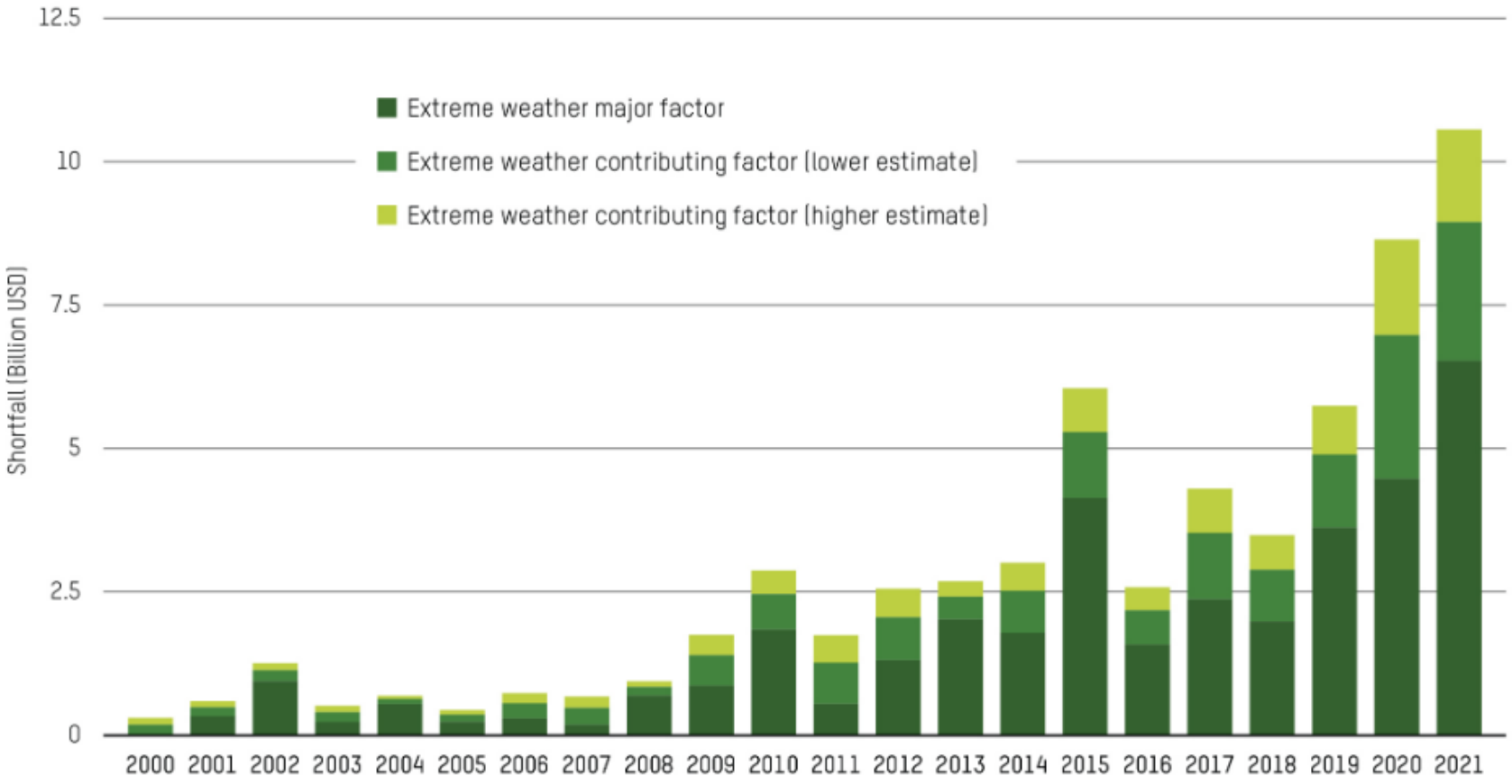
Figure 4: Funding shortfalls for extreme-weather-related UN appeals, 2000–2021



Notes: USD amounts are in 2021 real terms. The dark green sections cover appeals for which extreme weather was a major factor, and therefore 100% of the appeal shortfall is counted. The higher estimate (lightest green) counts 50% and the lower estimate (middle green) counts 30% of the value of the appeal shortfall.

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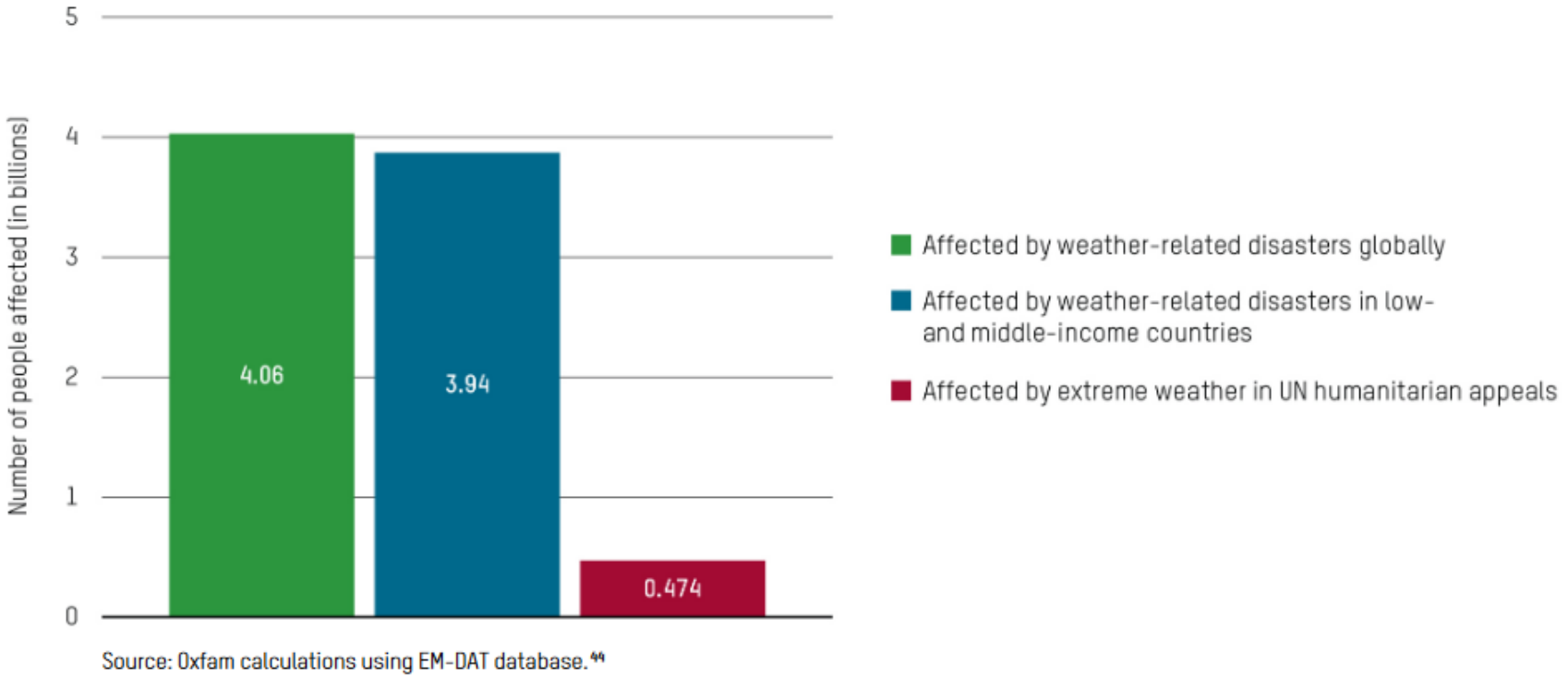
inadequacy of data
for mountain regions
hirisk.org

Figure 6: Number of people affected by weather-related disasters, 2000–2021



Source: Oxfam calculations using EM-DAT database.[44]

Loss and Da

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hirisk.org

Table 3: Estimated future costs of loss and damage

Source, year	Estimated costs
IFRC 'The cost of doing nothing' (2019)[59]	2030: humanitarian funding requirements for climate-related disasters could grow to $20bn per year.
Economist Intelligence Unit (2019)[60]	Measuring each country's direct exposure to loss from climate change and extreme weather events found that by 2050, GDP at risk could be: Africa (most vulnerable) 4.7% Latin America 3.8% Middle East 3.7% Eastern Europe 3% Asia Pacific 2.6% Western Europe 1.7% North America 1.1%
Markandya and González-Eguino (2018)[61]	Total annual residual damages for developing countries: 2030: $290–$580bn 2040: $551bn–$1tn 2050: $1.1–$1.7tn
Climate Analytics (2015)[62]	Projected macro-economic damage of climate change for developing countries: 2030: $400–$431bn 2050: $1–$1.8tn
Deloitte (2022)[63]	3°C trajectory could cost the U.S. economy $14.5 trillion by 2070.

Where do mountains come in?

Continuous attempts to bring *mountains* on the agenda since 1992

Less congruence in approach than SIDS (Small Island Developing States)

In some regions (like High Mountain Asia) complicated by political conflict

Often central governments themselves not located in mountains (perceived *marginal issue*) – leading to an increased emphasis of '*upstream-downstream linkages*' and '*cascading risks*'

Mountain Partnership (FAO)

Formal and government led

MOUNTAIN PARTNERSHIP STEERING COMMITTEE

The Steering Committee of the Mountain Partnership provides programmatic orientation to the Mountain Partnership, oversees the preparation of a four-year strategy and a two-year workplan. The 18-member body monitors the work of the Mountain Partnership Secretariat, the implementation of the 'Mountain Agenda' and the achievements and impacts of their respective electoral groups.

Steering Committee members are elected to four-year terms by the various membership categories and regions. They meet annually and select a Chair and Vice Chairs.

The following governments and organizations are serving on the 2022-2025 Steering Committee:

Donor Organization to the MPS
Italy (Chair)

Governments
Asia and the Pacific: Nepal
Europe: Andorra, Romania (Alternate)
Central Asia: Kyrgyzstan
Near East and North Africa: Yemen
North and Central America and the Caribbean: **Dominican Republic (Vice Chair)**, Guatemala (Alternate)
South America: Peru, Argentina (Alternate)
Sub-Saharan Africa: Malawi, Uganda (Alternate)

Host Institution of the MPS
Food and Agriculture Organization of the United Nations

Intergovernmental Organization
United Nations Environment Programme, United Nations University (Alternate)

- Our Vision and Mission
- Governance and Strategy
- **Mountain Partnership Steering Committee**
- Mountain Partnership Secretariat
- Logos and PowerPoint presentations
- FAQ
- Definitions

2024
- MP Steering Committee meeting report (22 February 2024)

2023
- MP Steering Committee meeting report (26 October 2023)
- MP Steering Committee meeting report (20 April 2023)

2022
- MP Steering Committee meeting report (1 December 2022)
- MP Steering Committee meeting report (29 September 2022)
- MP Steering Committee meeting report (30 August 2022)
- MP Steering Committee meeting report (21 June 2022)
- MP Steering Committee meeting report (30 March 2022)
- MP Steering Committee Documents (13 January 2022)

2021
- MP Steering Committee

Mountain Range Organisations

Alps: Alpine Convention - https://www.alpconv.org/en/

- Convention that includes a lawyer and proposes legal texts that are not binding but are used by arbitration courts in case of inter-governmental as well as country-internal disagreements

Carpathians: Carpathian Convention http://www.carpathianconvention.org/

- Similar to above, less developed/younger

Andes: CONDESAN https://condesan.org/

- Network of organisations working in the Andes (research heavy)

Hindukush-Himalaya: ICIMOD https://www.icimod.org/

- Intergovernmental organization (8 countries) that supports governments with expertise and provides platform for exchange

Mountain Research Initiative

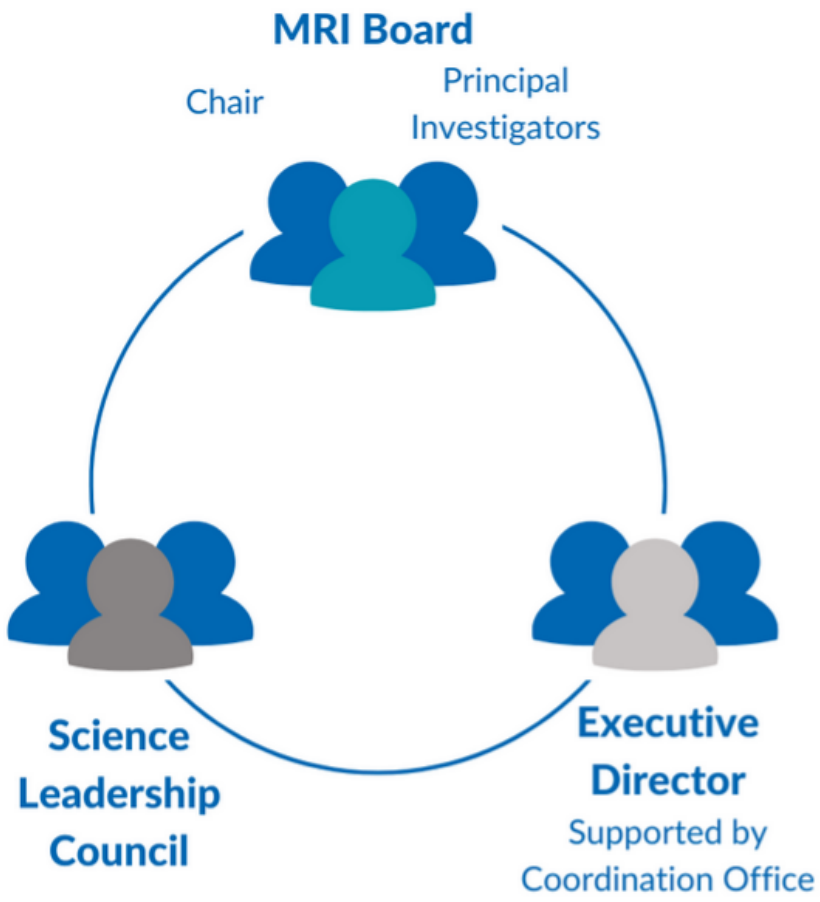
MRI Governance

The MRI Board (Chair and co-Principal Investigators), together with the MRI Science Leadership Council and the MRI Executive Director, constitute the MRI Governing Body. Together, they oversee and provide strategic direction and advice to support the implementation and development of both flagship and community-led activities, with the support of the Coordination Office staff.

As a team, we focus on supporting the mountain research community in making meaningful and fulfilling connections for our changing mountains and safeguarding them as social-ecological treasures for ecosystems and people worldwide.

Learn more about us:

- Principal Investigators
- Science Leadership Council
- Coordination Office



Adaptation at Altitude

Our Objective

The Adaptation at Altitude programme seeks to increase the resilience and adaptive capacity of mountain communities and ecosystems to climate change by:

- Improving the knowledge of appropriate climate change adaptation strategies in the mountains
- Transferring that knowledge through science–policy platforms to inform decision-making in national, regional and global policy processes

Adaptation at Altitude will foster exchange among the mountain areas of the world and seek short and long-term solutions to the problems arising from climate change.

International Mountain Conference

#imc25

Who is missing?

Course content

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How do international deliberations look like

From global (Conference of parties)…

… to regional (transboundary multilateral exchanges) …

… to local (transboundary bilateral exchanges)

Conference of Parties (UNFCCC)

1. Presidency and Bureau

Presidency: Each year, a member nation assumes the presidency to host and shape the priorities of the conference.

Bureau: The COP Bureau consists of elected members

2. Plenary Sessions

Opening Plenary: Delegates from nearly every country gather in a large hall to agree on the agenda and officially launch the work ahead.

Closing Plenary: The final session where all parties review texts, hear statements, and adopt formal decisions.

3. Negotiating Groups

Contact Groups: Formal meetings where negotiators tackle specific agenda items by drafting and revising text paragraph by paragraph.

Informal Consultations: Smaller, focused meetings designed to resolve specific disputes or reach consensus on text before taking it to the contact group.

High-Level Segments: Reserved for ministers and heads of state to deliver national statements and make key political decisions.

COP28 signals beginning of the end of the fossil fuel era

In a demonstration of global cooperation, negotiators from nearly 200 countries for the first time recognized the need to transition away from fossil fuels. “Whilst we didn’t turn the page on the fossil fuel era in Dubai, this outcome is the beginning of the end,” said UN Climate Change Executive Secretary Simon Stiell in his closing speech. UN chief António Guterres emphasized that the era of fossil fuels must end with justice and equity. “Developing countries must be supported every step of the way,” he said in his statement on the closing of COP28.

COP 28 concluded with a decision on the outcome of the first ‘global stocktake’ of the world’s efforts to address climate change under the Paris Agreement.

Find out more about what was achieved at COP28.

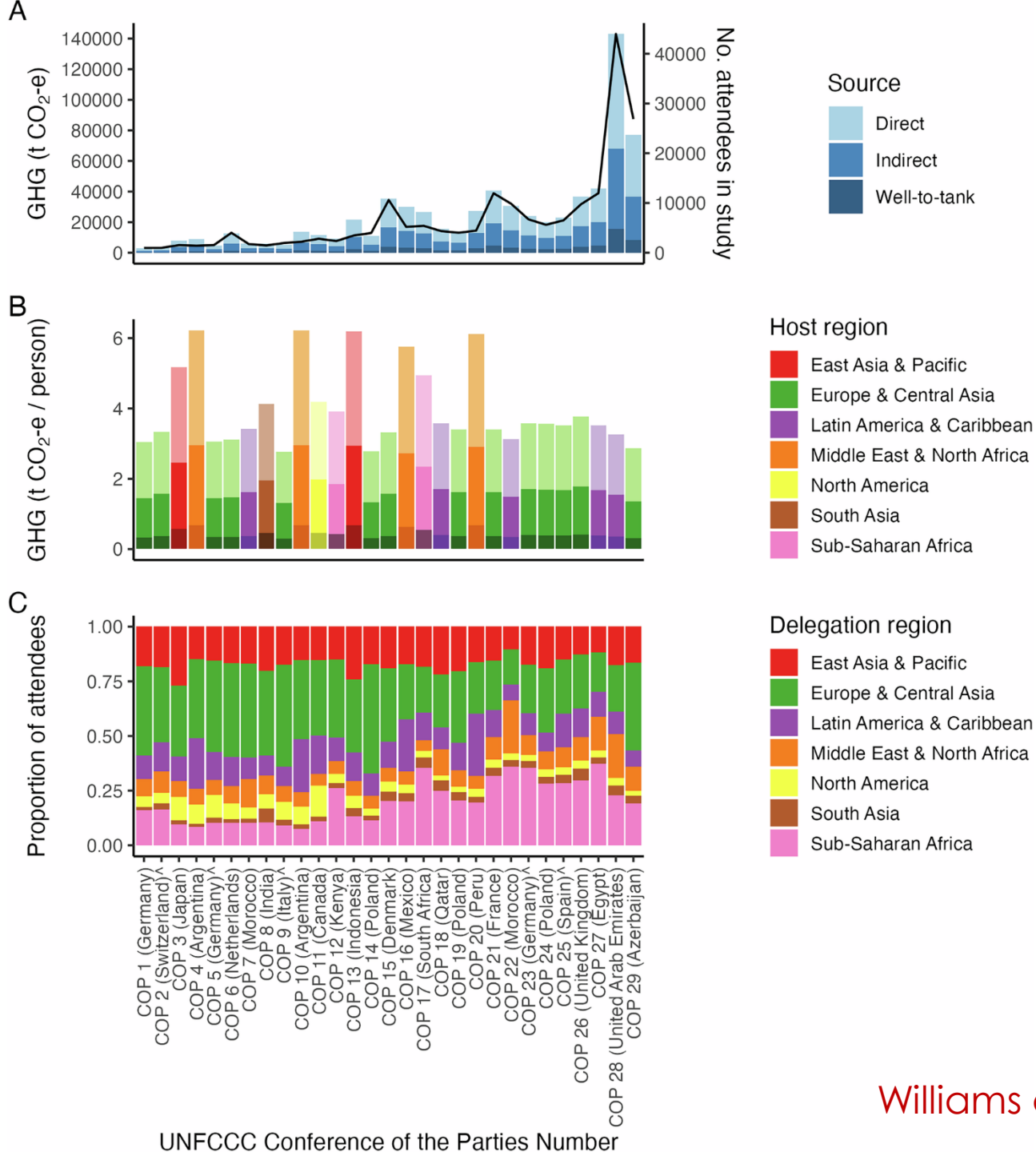
COP29 concludes with climate finance deal

The UN climate conference in Baku, Azerbaijan, concluded on 24 November with an agreement calling on developed countries to deliver at least $300 billion per year to developing countries by 2035 to drastically reduce greenhouse gas emissions and protect lives and livelihoods from the worsening impacts of climate change. “I had hoped for a more ambitious outcome – on both finance and mitigation – to meet the great challenge we face,” UN Secretary-General António Guterres said in his statement on COP29. “But this agreement provides a base on which to build.”

COP30 closes with agreement to step up support for developing countries

The UN climate conference in Belém, Brazil, concluded on 22 November with an agreement calling for a tripling of funding for developing nations to protect their people from the growing impacts of the climate crisis, and for stepping up support for workers and communities in the transition to clean energy. "COP30 has delivered progress," UN Secretary-General António Guterres said in his statement on COP30. Yet, "the gap between where we are and what science demands remains dangerously wide." Keeping global temperature rise below 1.5°C by the end of the century is vital, and requires "deep, rapid emission cuts – with clear and credible plans to transition away from fossil fuels and towards clean energy,” he stressed.

The outcome texts - the Belém Political Package - can be found here.



Williams et al. (2026)

Regional deliberations – the case of High Mountain Asia

ICIMOD

About us | Current work | Media | Past work | Our resources | Stay informed

BACK TO EVENTS

MEETING

The Hindu Kush Himalaya Ministerial Mountain Summit 2020

Programmes
ICIMOD

Venue
Online (MS Teams)

Date & Time
15 October 2020

Contact
Udayan Mishra & Bhawana

HINDU KUSH HIMALAYA REGIONAL MEMBER COUNTRIES

AFGHANISTAN BANGLADESH BHUTAN CHINA INDIA MYANMAR NEPAL PAKISTAN

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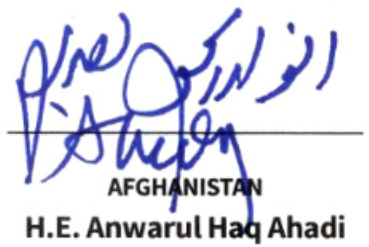
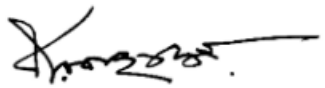
Ministerial Declaration on the HKH Call to Action

We, the Ministers, agree:

- to hold a biennial HKH Ministerial Mountain Summit to take joint actions for strengthening regional cooperation in the HKH and to review and discuss emerging issues;
- to promote a united voice for the HKH at regional, global and UN platforms;
- to hold a multi-stakeholder HKH Science-Policy Forum on a regular basis in order to enhance the uptake of scientific evidence for improving policies in the region focusing on mountain environments and livelihoods;
- to constitute a Task Force with high-level representation from the eight HKH countries to assess the feasibility of establishing a regional institutional mechanism. The Task Force should assess similar regional collaborative platforms/institutions from different parts of the world and recommend a feasible configuration for the HKH region in a report to be submitted to the next HKH Ministerial Summit. The Task Force would also monitor the progress of the "HKH Call to Action".



AFGHANISTAN
H.E. Anwarul Haq Ahadi
Minister, Ministry of Agriculture, Irrigation and Livestock



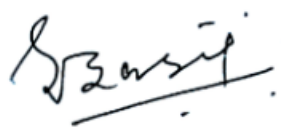
BANGLADESH
H.E. Bir Bahadur Ushwe Sing MP
Minister, Ministry of Chittagong Hill Tracts Affairs

BHUTAN
H.E. Lyonpo Yeshey Penjor
Minister, Ministry of Agriculture and Forests

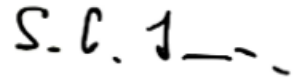
CHINA
H.E. Zhang Yaping
Vice President, Chinese Academy of Sciences

INDIA
H.E. Prakash Javadekar
Union Minister, Ministry of Environment, Forests and Climate Change

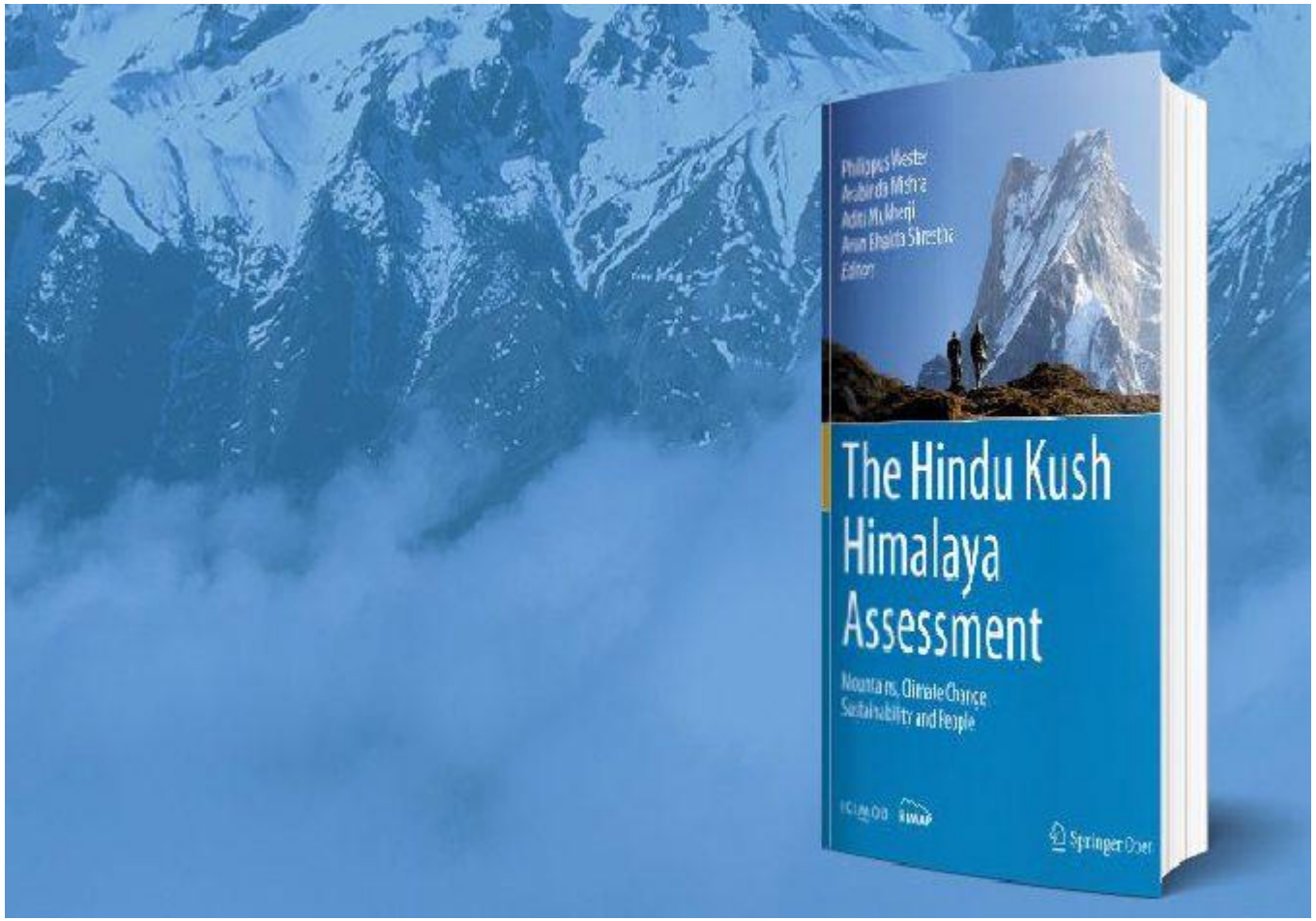
MYANMAR
H.E. U Ohn Winn
Union Minister, Ministry of Natural Resources and Environmental Conservation



NEPAL
H.E. Shakti Bahadur Basnet
Minister, Ministry of Forests and Environment



PAKISTAN
H.E. Syed Fakhar Imam
Federal Minister, Ministry of National Food Security and Research



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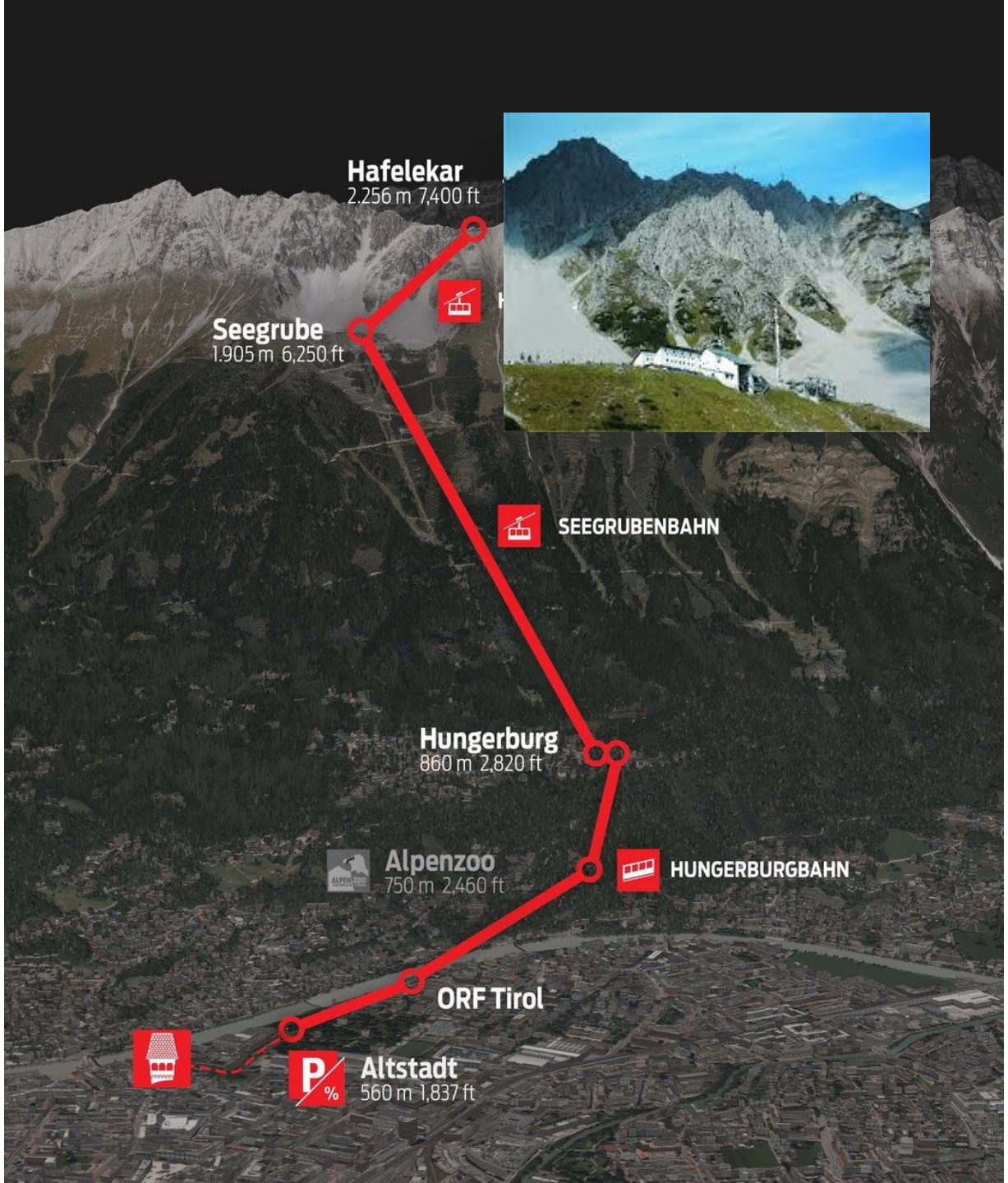
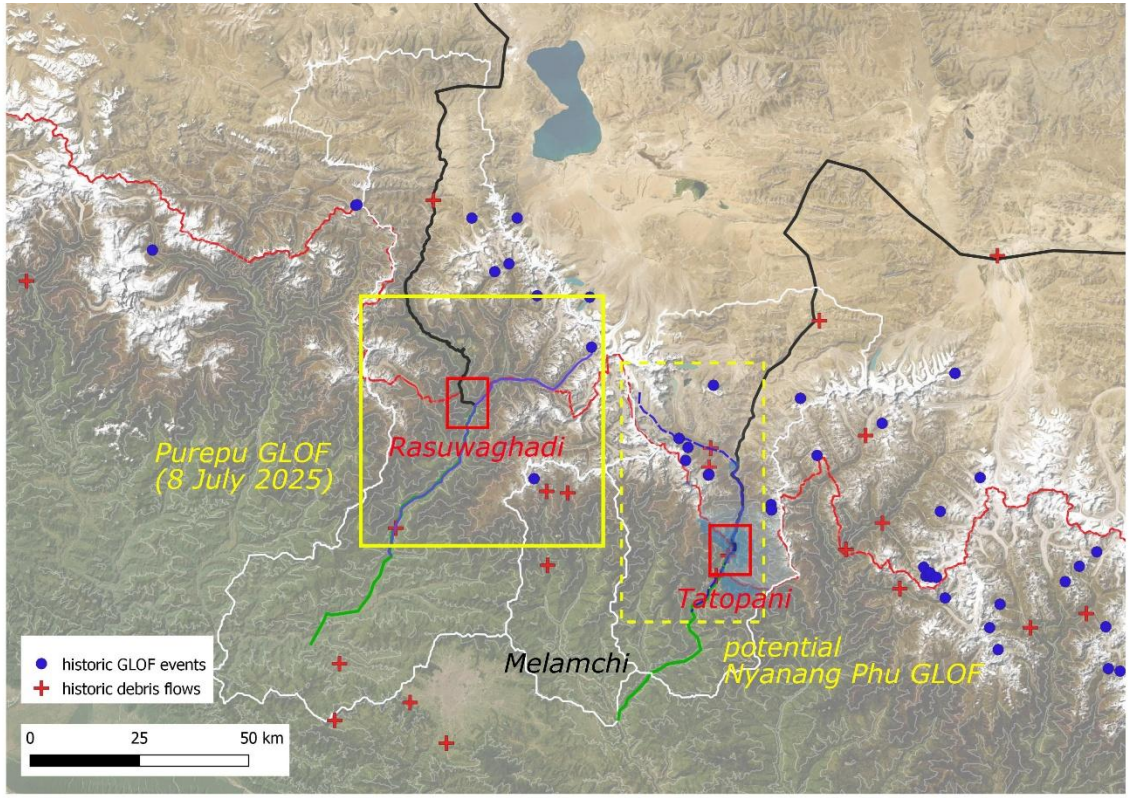
Water, ice, society, and ecosystems in the Hindu Kush Himalaya

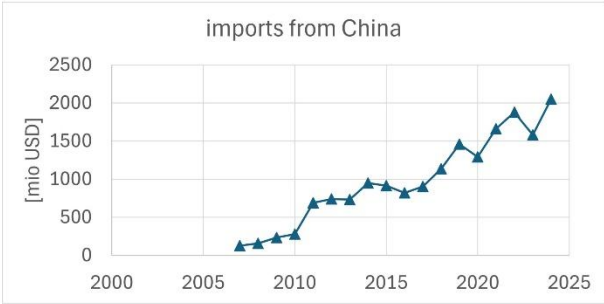
An outlook

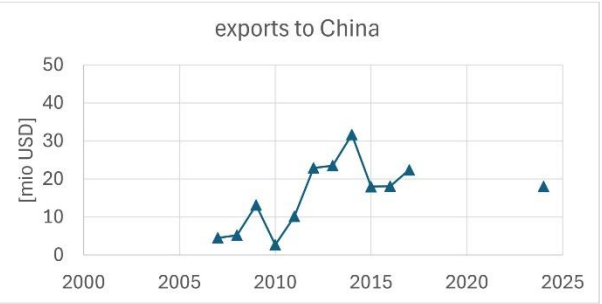
2022 – learning visit to Alpine Convention and deliberations for a next step in the region towards binding agreements

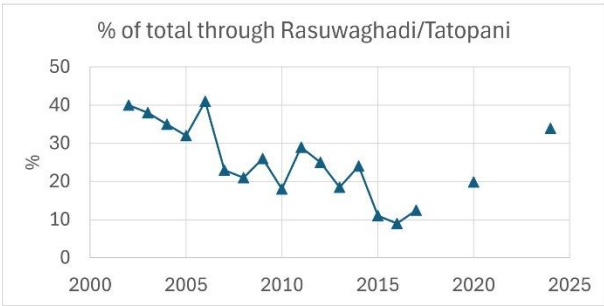


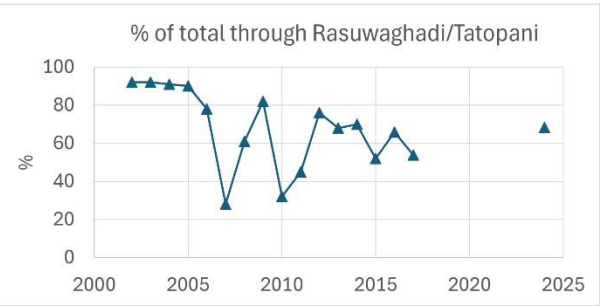
Local deliberations



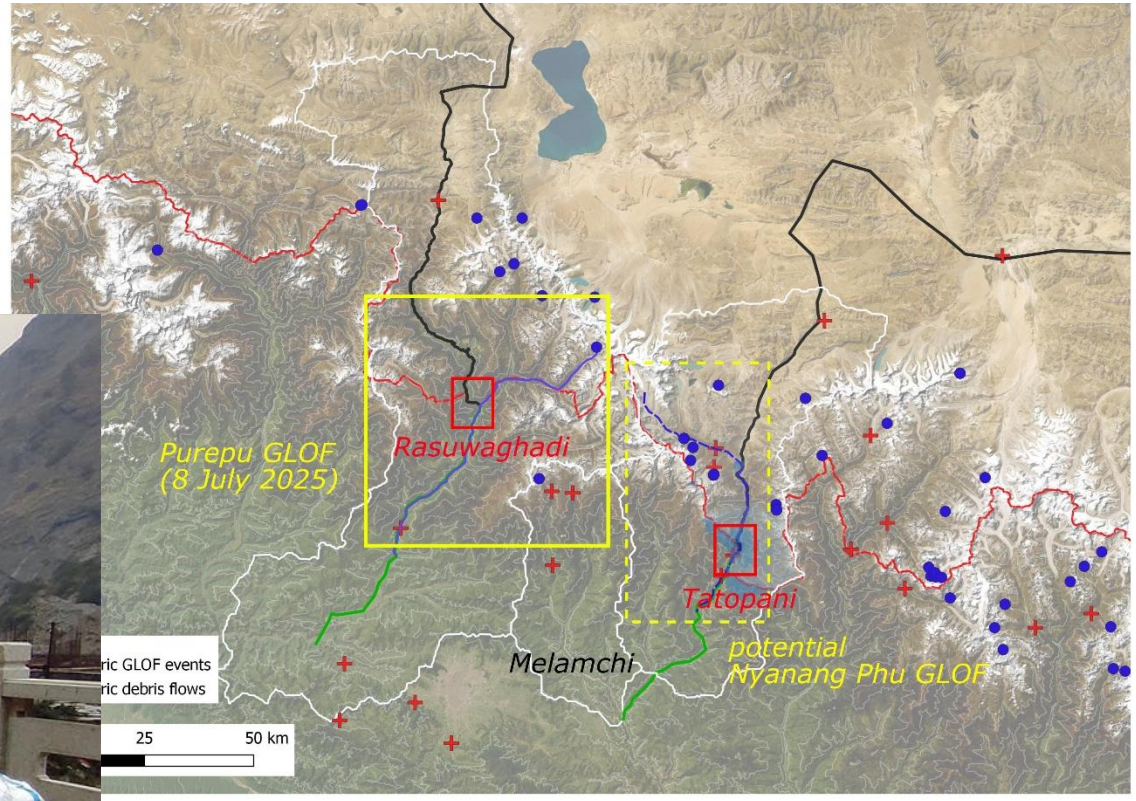


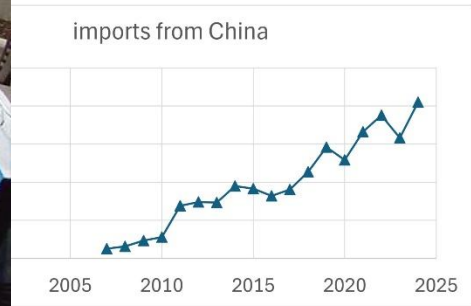


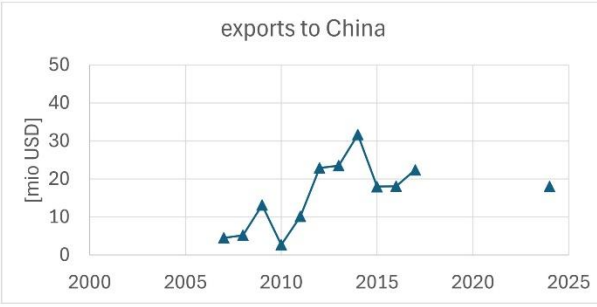


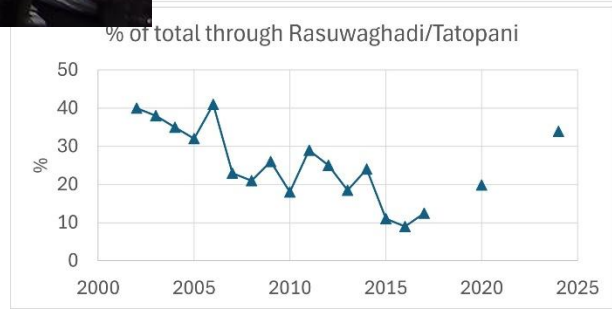


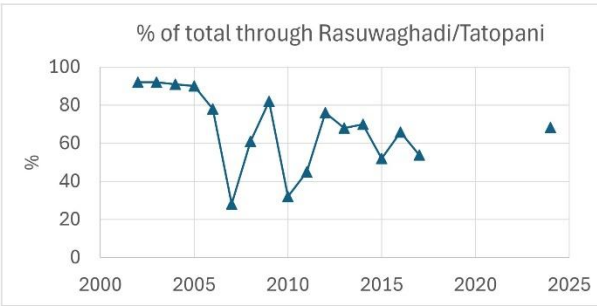
Local deliberations











RHA Report No. CN1

Chinese (p.1) / Nepali (p.5) /
English (p.9)

灾害快速评估报告

地点：中国西藏日喀则吉隆县萨勒乡，Purepu冰川冰面湖，吉隆藏布支流（Bhote Koshi）

国家：中国（跨境至尼泊尔）

纬度：28.402561°N — 经度：85.646705°E — **海拔：5160米**

日期：2025年7月7日至8日

当地时间：未知（首次报告的影响时间为0530中国标准时间/0315尼泊尔标准时间）

灾害类型：冰川湖溃决洪水（GLOF）

影像资料：（来源：Sentinel-2，2024年6月至2025年7月）

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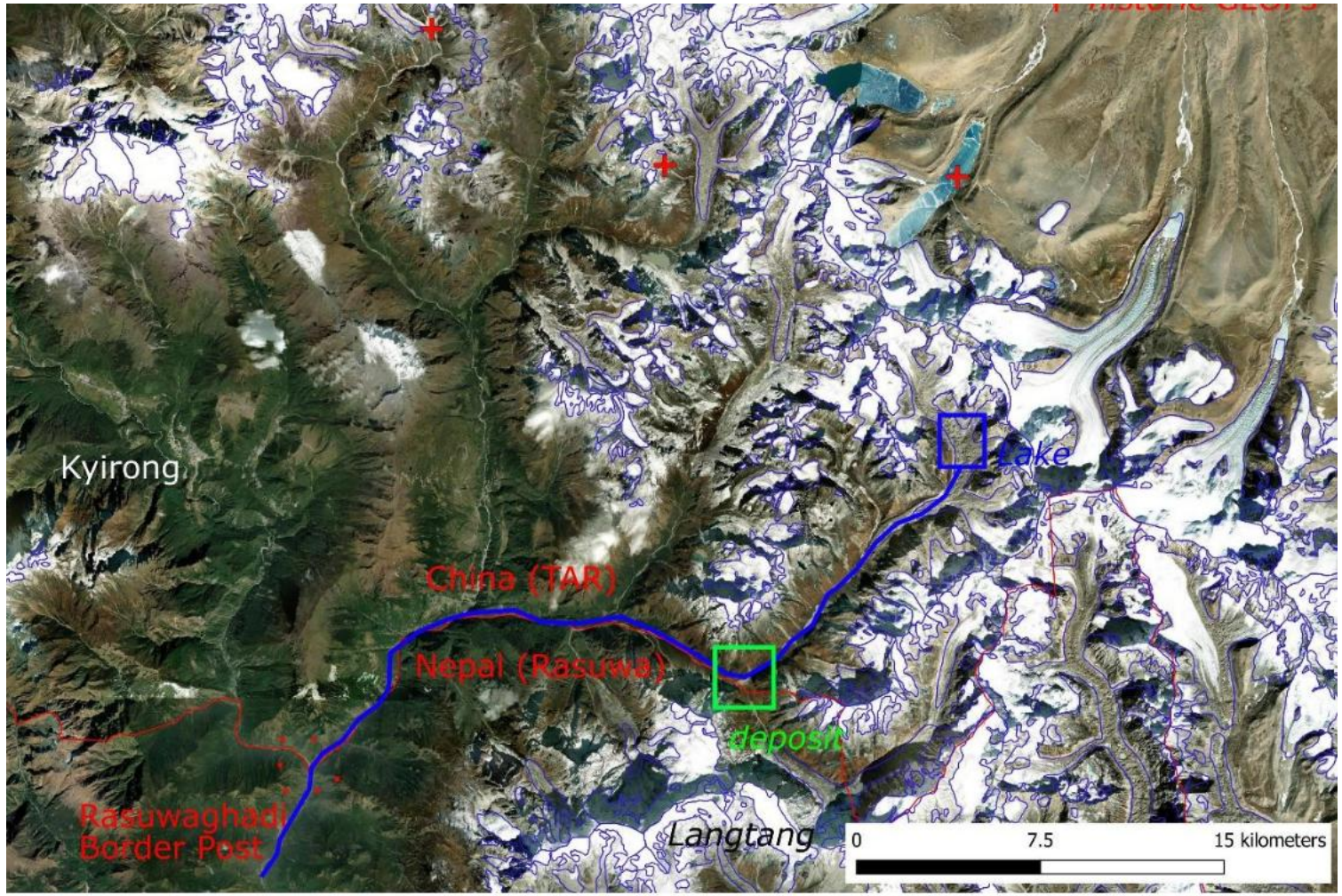
https://hirisk.org/risk-assessments/



https://hirisk.org/risk-assessments/

Bilateral Deliberations

Find common ground in focus groups:

Academic

International Organizations and ministers/Civil Society

Defend those arguments towards the other parties

Present a common agreement

A	B	C
	Role	Student Name
China	Ministry	
	Business (cars)	
	Business (construction)	
	Academia	Ekaterina Fedoseeva
Nepal	Civil Society	Laura Reutter
	Minister	
	Academia (Fores	Valerie Kalt
	Academia (Clima	Rachel Lecroix
	International Age	Xuan Han Chen
Pakistan	Civil Society	
	Business	Sebastian Buțu
	Ministry	
	Academia	David Fulks
	International Organiation	

COP is officially over – you're asked by your media to report success or failure

Did you achieve what you wanted to get through? Where did you need to compromise?

Is there self criticism for your own country, your own interest group?

Did you see potential to find global common ground for interests of mountain regions?

How would you change your strategy for next time?

What stakeholder need to be brought in for next deliberations?

COP is officially over – you're back to your Self

COP is officially over – you're back to your Self

Could you find agreements with others? What made coming to agreements or strategies most difficult? What were you lacking to come to agreements?

Is virtual exchange useful? Do you need to see the others in person, do you need to travel to their countries?

Do you think it is realistic to come to agreements in such a setting and what would it take that they are being adhered to?

Course content

28 May (0900 – 1500)

09:00 – 09:45	Disaster Risk Reduction
09:45 – 10:00	Break
10:00 – 10:45	Concrete recent examples
10:45 – 11:00	Break
11:00 – 12:00	International Platforms to address Mountain Risks
12:00 - 13:00	Lunch Break
13:00 - 13:45	Intro to the Serious Game – how do international deliberations look like?
13:45 - 14:00	Break
14:00 - 15:00	new data and method avenues

Course content

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The future of early warning

Impact based forecasting (https://www.anticipation-hub.org/learn/emerging-topics/impact-based-forecasting)

- ‘What will the weather do’ rather than ‘What will the weather be’
- What is useful for people?

Government of Pakistan
Ministry of Defence (Defence Division)
Pakistan Meteorological Department
Research and Development Division
Islamabad, Pakistan

GLOF ALERT

GLOF/Met-3(2)/2025/

Dated: 02nd April 2026
Issue Time: 03:00 PM

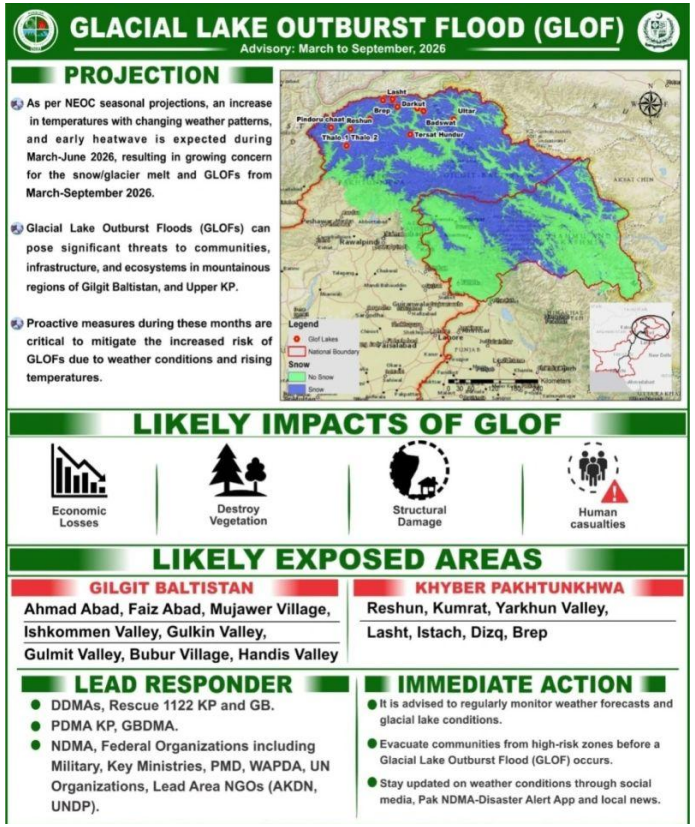
A westerly wave is likely to affect upper parts of the country from the night of 02nd April and is expected to persist till 08th April with intermittent rain. Under the influence of this weather system, widespread rain/thunderstorm and gust, accompanied by moderate to heavy rainfall and snowfall over hill tops, is expected in Mankiyal, Matiltan, Kumrat and Utror-gabral valleys of Swat, and Kandia valley of lower Khyber Pakhtunkhwa (KPK) and Resun-Booni, Arkari and Madaklasht of Chitral upper KPK. The districts of Gilgit Baltistan (GB) Ghizer, Hunza, Nagar, Astore, Guanche, Diamer, Kharmong, Skardu and Shigar are likely to be affected. The expected persistent precipitation, coupled with the snowmelt in high-altitude catchments, and intense runoff generation may significantly aggravate hydro meteorological hazards in vulnerable valleys.

Likely Weather-Induced Impacts:

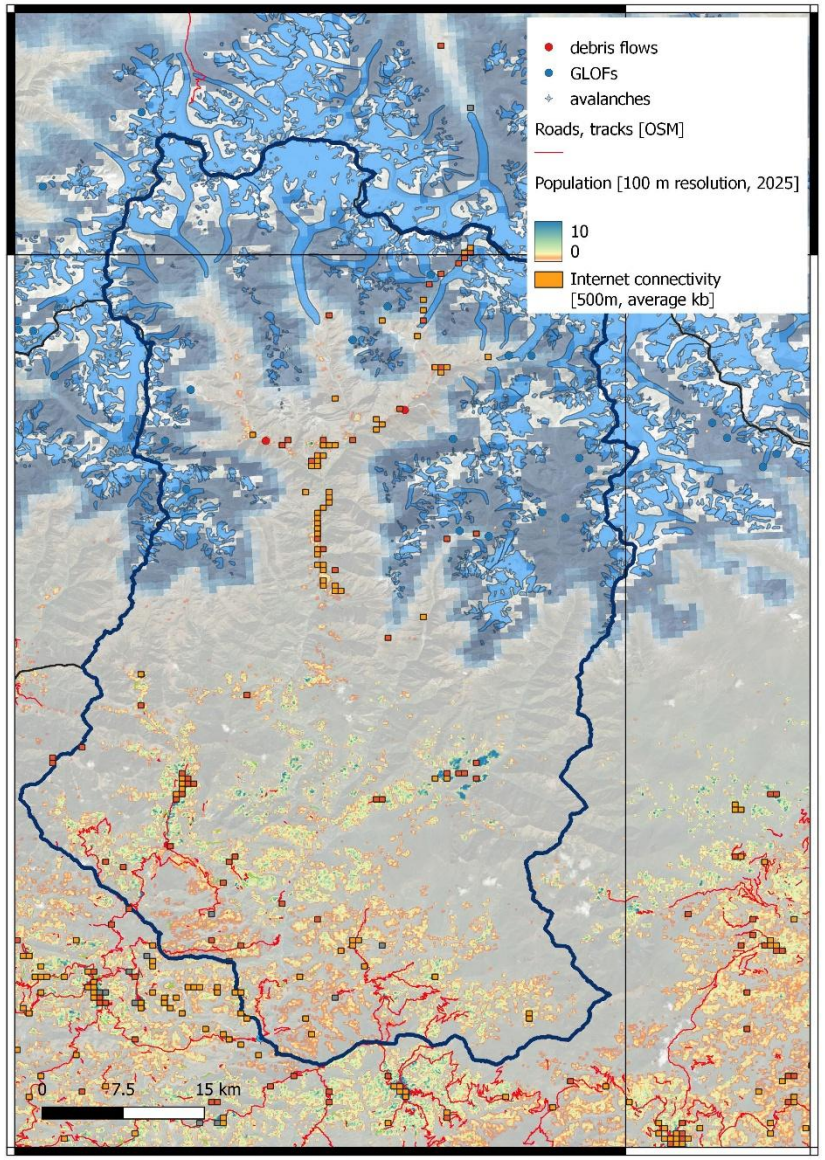
- Probability of Glacial Lake Outburst Flood (GLOF) in vulnerable glaciated basins and downstream settlements.
- Flash flooding is likely in local streams, nullahs and tributaries particularly in KPK river basins.
- Heavy rainfall over steep slopes may trigger landslides, mudslides, debris flow in mountainous districts of KP and GB.
- Mountain roads and communication links may face temporary closure/blockage due to debris flow and flash runoff.

2. All concerned are advised to remain alert and take necessary measures to avoid any untoward situation.

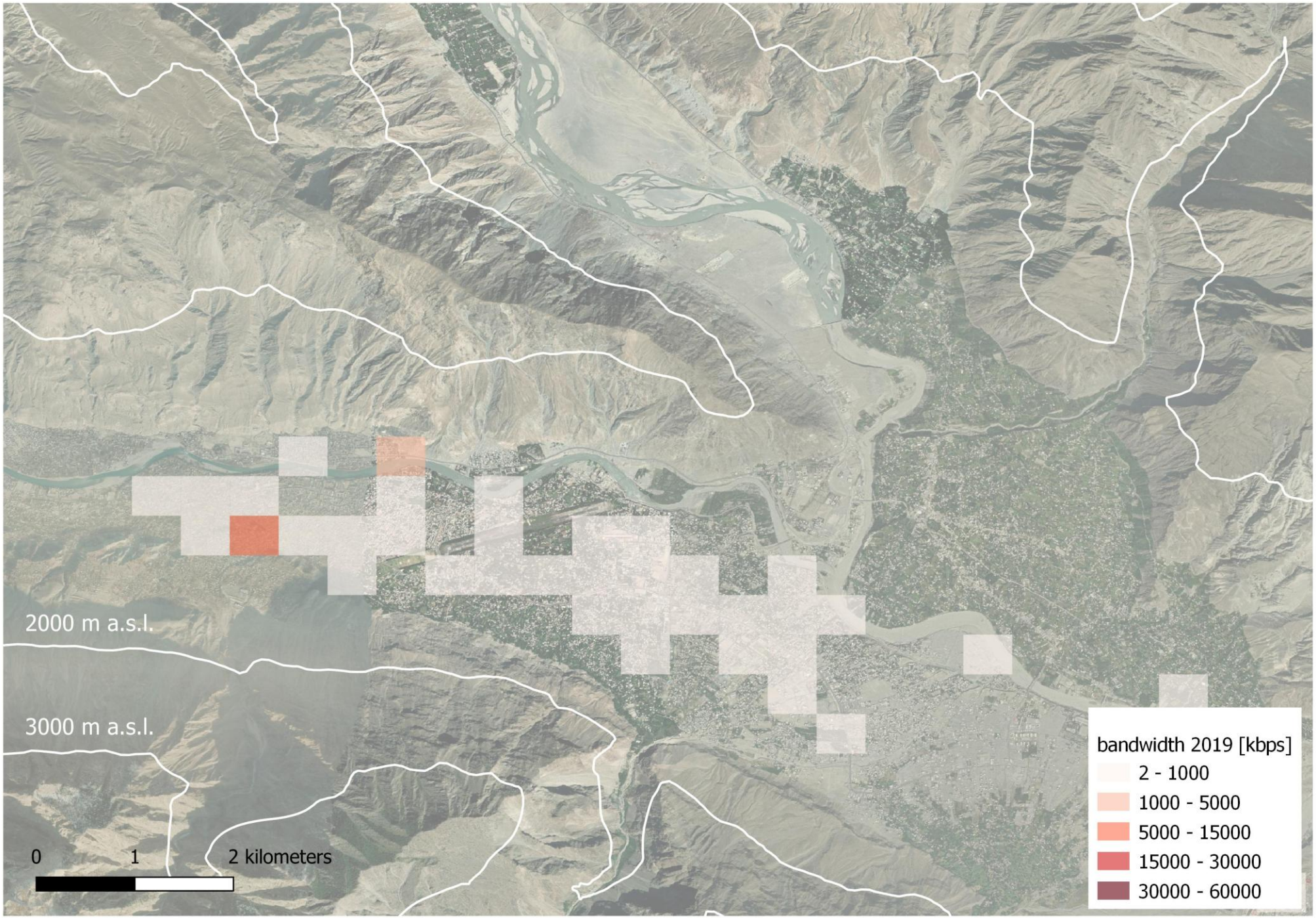
(Zeenat Yasmeen)
Meteorologist,
For Director General, PMD



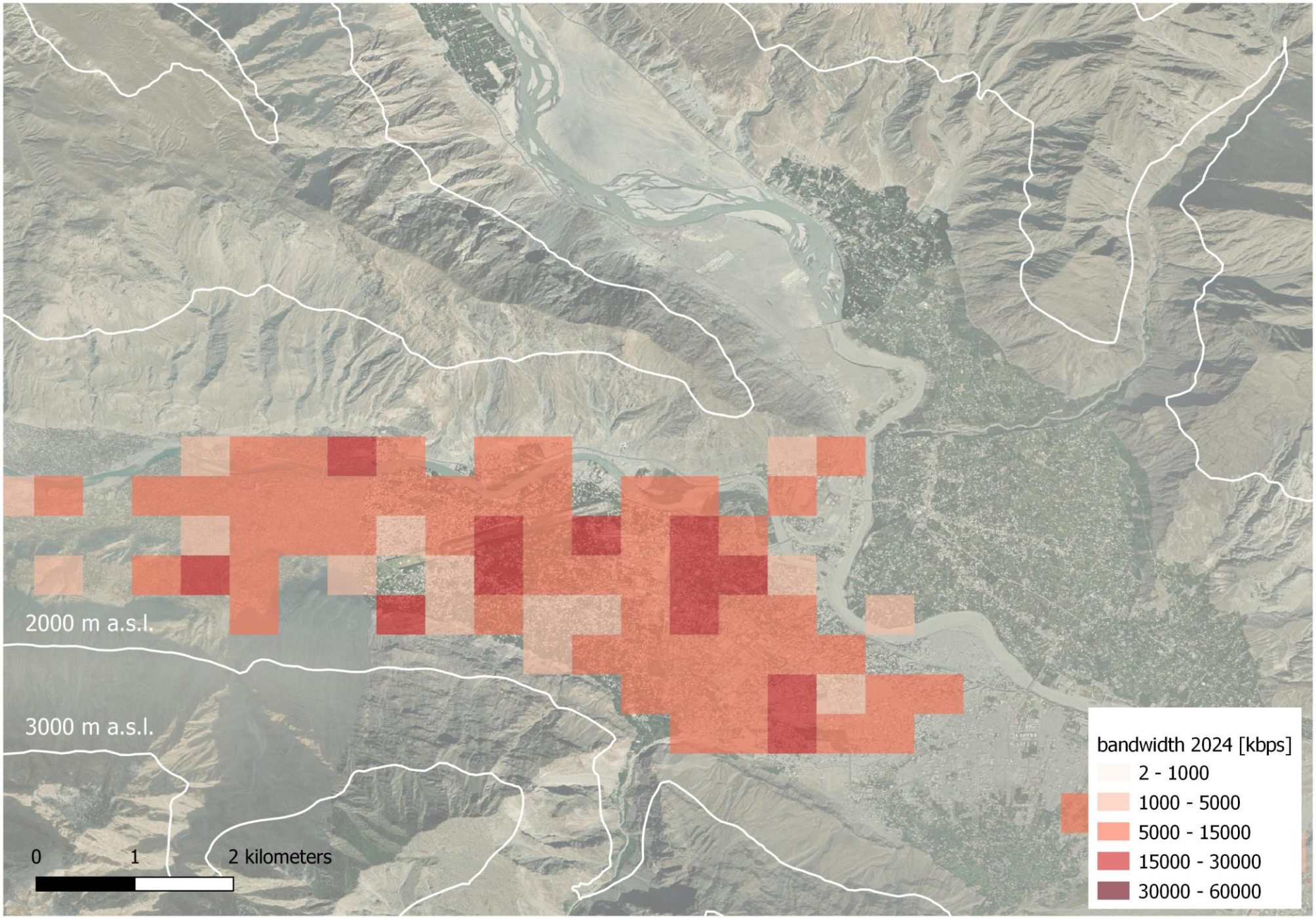
The future of tracking data access



Ookla data



Internet connectivity
Ookla 2019



Internet connectivity
Ookla 2024

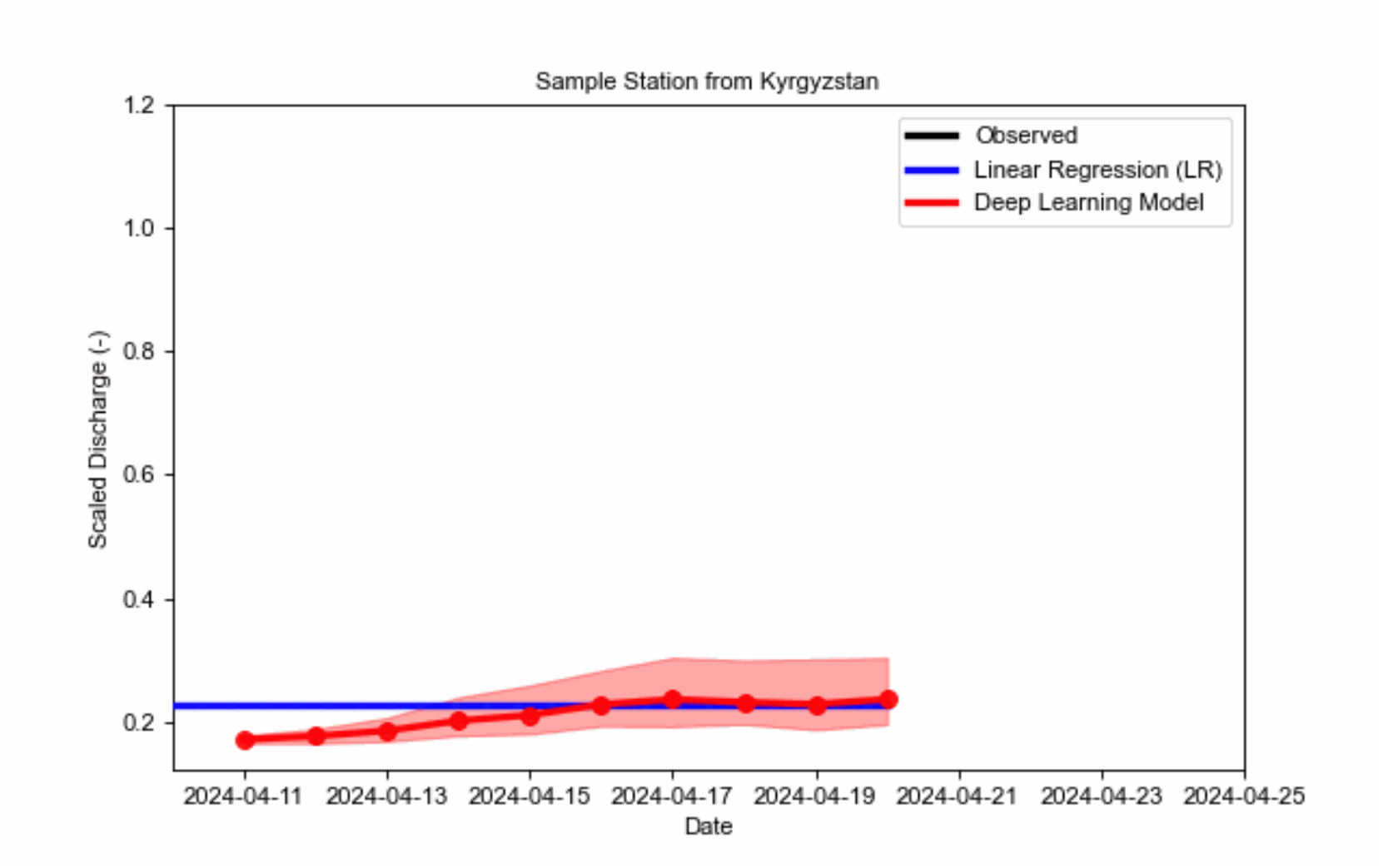
From conventional to the future of forecasting

Figure 1: Traditional water level measurements are carried out manually in Central Asia twice per day. These traditional workflows are gradually replaced by modern montioring infrastructure that measures many relevant variables at high-frequency. These creates a data flood overwhelming the local institutions. Source: hydrosolutions GmbH.

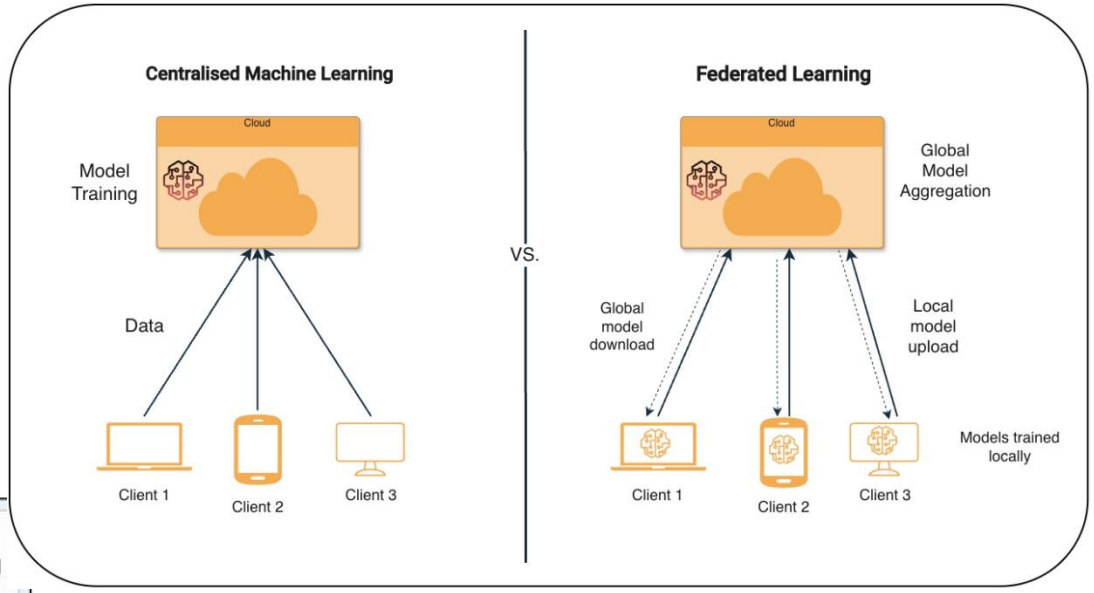
How do we bring conventional ways of monitoring into the age of Big Data, ML and AI?

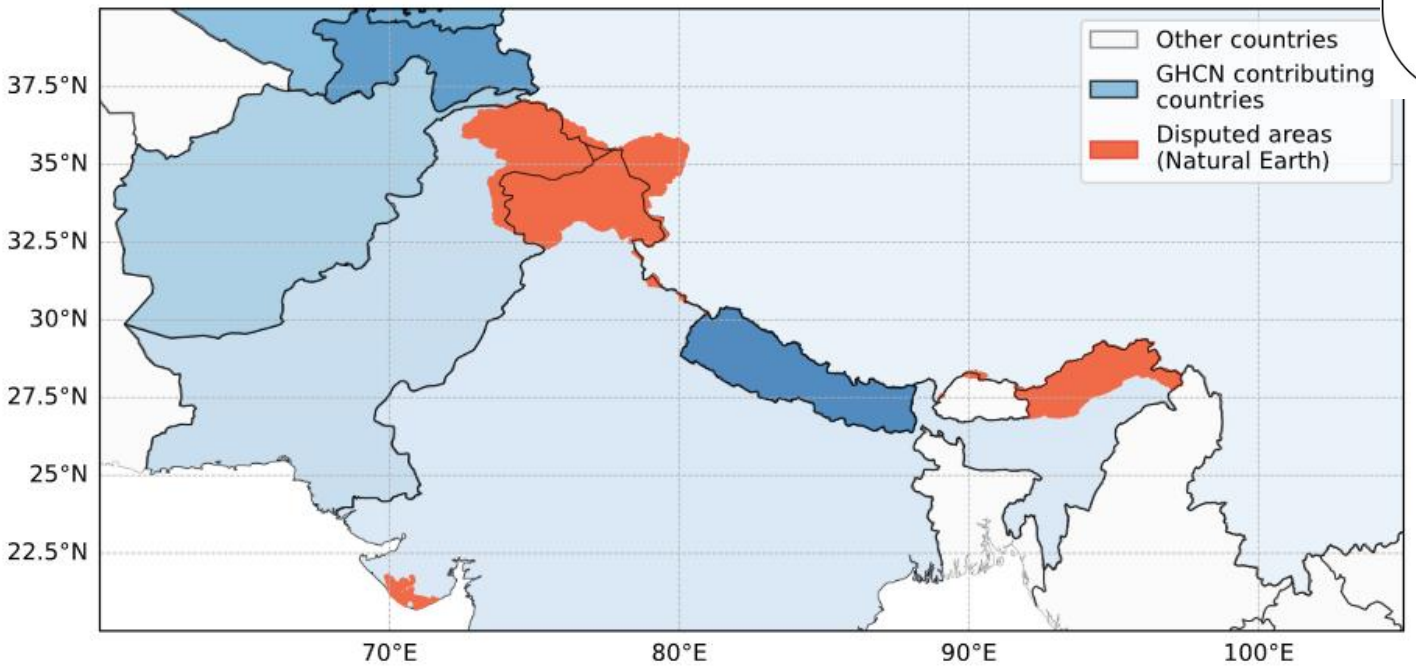
https://www.hydrosolutions.ch/

From conventional to the future of forecasting

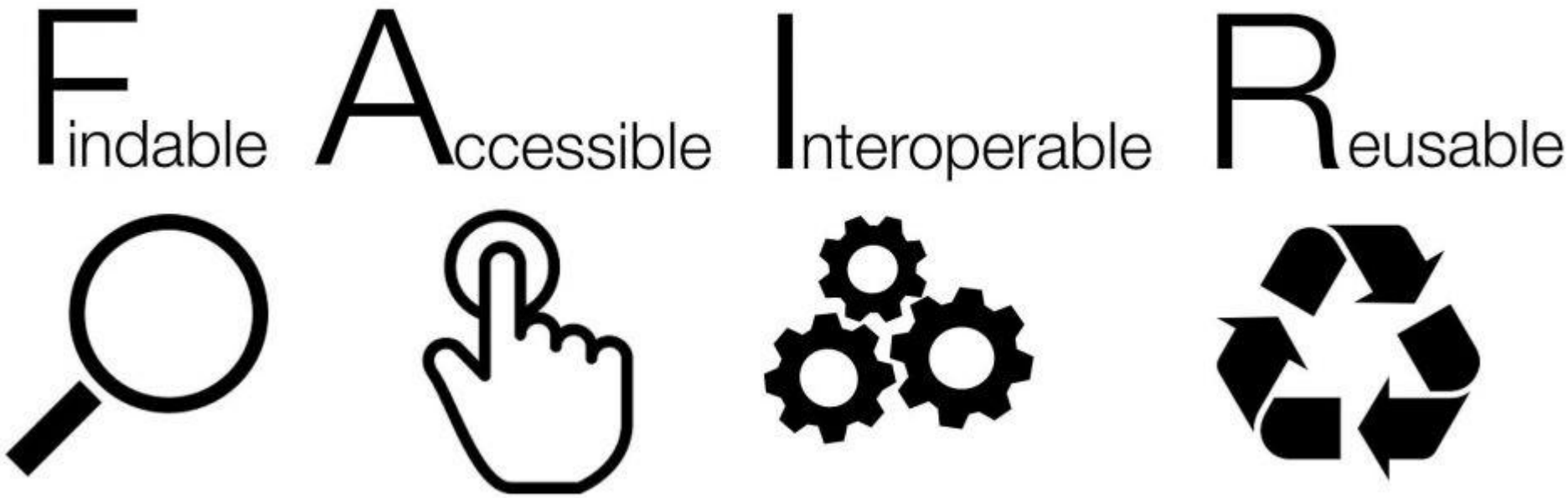


Federated Learning – how to model when countries don't share?





Is ‘FAIR Data’ enough?



CARE data

https://www.gida-global.org/care-principles-copy